W O R L D   M E T E O R O L O G I C A L   O R G A N I Z A T I O N

RGB EXPERTS AND DEVELOPERS WORKSHOP

1-3 APRIL 2025

NORRKÖPING, SWEDEN

FINAL REPORT

SUMMARY OF RECOMMENDATIONS

General Recommendations:

1. The group emphasised the importance of training, as more RGB products have become available. Quick guides are very useful in this regard.
2. The use case (professional, public outreach) of each RGB product should be made clear.
3. The group recommended to consider users with colour vision deficiency when and allow adjustments, when creating and implementing RGBs.
4. The group recommended structuring the RGBs in four categories: 1) Basic, 2) Special Applications, 3) RGBs under development. and 4) Legacy RGB composites. It also made recommendations on Other visualisations such as the GeoColour composite.
5. A distinction should be made between professional RGB products, and RGB/imagery products for public relations use.
6. As per the recommendations from the 2022 workshop and discussions during this workshop, mapping application areas against relevant RGB composites is beneficial to operational users.
7. It is recommendable to share this workshop report in wider relevant networks.
8. It is recommendable to invite more operational experts to future workshops.
9. The group recommends organizing an annual online meeting to follow up on the developments of RGB products as discussed in this workshop.

Specific RGB recipe recommendations:

1. A RGB recipe standard should at least encompass: channels, ranges, gamma factors; sun zenith angle correction, Rayleigh correction.
2. On the precision of ranges, a maximum of one decimal digit should be used, and rounding to full integers for brightness temperatures.
3. The GeoColour visualisation requires improvement in the day/night transition (low cloud detection).

4. The Flexible Combined Imager (FCI) on MTG generates for some channels data simultaneously at two different spatial resolutions, e.g., IR10.5 at 1km high resolution, and at 2km normal resolution. When combining such channel data with other channels within one colour beam, to avoid artefacts, it is recommended to use imagery at identical spatial resolution, e.g., to use IR10.5 2km data in combination with IR12.3 2km data.
5. RGBs should contain "Day" or "Night" in the product name full title when the product is available or usable only during day or night.
6. When creating moisture products using the 0.9-micron channel, the channel ratio 0.9/0.8 should be used instead of subtracting the channels.
7. The group recommends renaming SO2 RGB product to Volcanic Emissions (SO2) RGB.
8. Many RGBs have a shorthand naming in use for simplicity. Recommendable to use the full names in guides and documentation and add a comment (This is often referred to as xyz RGB). The list of shorthand names is available in the RGB recipes document (Annex 3).
9. The Day Cloud Phase Distinction RGB should be considered as a variant of Cloud Type RGB and rename it to Day Cloud Type Distinction RGB (with alternative name being Day Cloud Phase Distinction RGB given widespread use of this name in use with ABI, AHI)
10. Use the Li & Shibata (2006) correction method as standard for solar zenith angle correction, for better results at high solar zenith angles near the terminator.

BACKGROUND

Workshop objectives:

Consolidate contemporary GEO-ring RGB composite standards and publish them in a workshop Report

- Identify novel RGB products
- Demonstrate applications RGBs using new-generation imagers (e.g. FCI)
- Identification of common platforms to share the RGB recipes and related material

All presentations provided at the event are available at:
https://wmoomm.sharepoint.com/:f:/s/wmocpdb/EgMfpeu2VatAlwpb3AplkvsBRpUP-bgi8LB8rmZGENfyUw?e=ZnnH62

INTRODUCTION

Welcoming Remarks and Logistics

The co-chairs Heikki Pohjola (WMO) and Martin Raspaud (SMHI) opened the meeting and welcomed participants. Participants introduced themselves in a tour-de-table (see Annex I for list).

Purpose and Scope of Meeting: Setting the scene (S Bojinski, H Pohjola)

S Bojinski recalled the objectives of the workshop:
- set standards for GEO spectral imagers RGB recipes, for forecasters, manufacturers
- discuss applications of RGBs and use cases

The workshop addresses a strong, global community effort to support forecaster use of RGB composites, using the full temporal and spatial information content of spectral imagers, with focus on the GEO ring. This started with defining RGBs from SEVIRI in 2003, continued with AHI over Asia-Oceania (2015), ABI over the Americas (2017) and now FCI over Europe and Africa (2024).

Standards are efficient to inform manufacturers, trainers, forecasters in weather offices, and viewers such as EUMETView, ePort. Common nomenclature, recipes, and training material help foster a global effort in applying RGBs.

The workshop is follow-up to the series of RGB expert and developer workshops in Boulder (2007), Seeheim (2012), Tokyo (2017), and Ft Collins (2022).

H Pohjola presented the WMO framework, the Space Programme role, and introduced regulatory material, and the history of RGB workshops. Goal is to produce a **WMO Guideline** based on the outcome of the workshop. WMO Guides are non-regulatory documents with recommended practices. They define practices, procedures and specifications; Members are invited to follow or implement; they help in developing meteorological and hydrological services in Members' respective countries.

B Connell recalled WMO Education and Training Programme (ETR) use of the term "frameworks" when referring to skills and knowledge, rather than "standards".

J Strandgren remarked that the group should agree on what should be standardised (channels, ranges, gamma factors; sun zenith angle correction, Rayleigh correction).

M Raspaud noted that local variants will remain important, such as for limb effect corrections.

Overview of RGB composite recipes, applications, limitations (Maria Putsay)

Maria Putsay presented document RGB_recipes _4_RGBWorkshop2025.docx. The online document was commented/edited by the workshop participants before the workshop, and the final result based on workshop discussions is included in Annex III. Several topics to be resolved in the workshop were presented. Europe and U.S. groups sometimes use different definitions of gamma factor. Also, the different naming conventions and RGB variants between the different instruments were introduced.

What about names in French or Spanish, or any other language? Meteo-France use English names, SNM Argentina is translating names into Spanish. DWD translate some of their RGBs into German. To be discussed how to proceed with the names in other languages than in English.

Based on a remark by I Al Abdulsalam about the precision of ranges, the group recommended to use a maximum of one digit after the comma, and round to full integers for brightness temperatures.

B Connell suggested to drop the term "inverted" in the recipe tables and just use the range. This avoids confusion as to whether the recipe is already inverted or needs to be inverted.

Recommendation: Add image examples for each RGB in the Guide.

NEW APPLICATION FINDINGS FOR ESTABLISHED RGBs

Regional use of Himawari RGBs (Bodo Zeschke)

Bodo Zeschke from Australian BoM and VLab perspective in WMO RA V (Asia-Oceania). GEO imagers of CMA, JMA, KMA and U.S. contributing to monitoring RA V.

Panther viewer software in BoM: Night Microphysics RGB with largest positive impact among BoM staff, then True Colour, Day Microphysics, Day Convection, Natural Colour RGB and Airmass RGB. Volcanic Ash RGB used only by subset of forecasters. He showed the example of monitoring the Lewotobi volcano, Indonsia, from Darwin VAAC using Volcanic Ash RGB, in addition to 3.9um and 8.7um enhanced imagery from the VOLCAT algorithm.

He presented several AHI-based RGB composites as adapted by BoM or RAV colleagues (the recipes are not known). Some are unique to RA V or BoM, and not reflected in the tunings for AHI.

He expanded on the development for RGB composites for colour-impaired forecasters, using the Night Microphysics RGB over southern Australia as an example. Contours are used to delineate areas of interest (e.g., cyan). It is not clear how these lines are generated (unrelated to warning areas, or likelihood of lightning as provided by LightningCast).

The question was raised how to include local characterization into RGB documentation? It is a difficult question that should be tackled by the workshop. (There can be local variants, but we need a default for each sensor, perhaps with mid-latitude and tropical variants).

He confirmed that Night Microphysics and Day Microphysics RGBs are both very popular developed already years ago.

RGB applications at CMA (Jian Lu)

Jian Lu (CMA) introduced the GEO imagers operated by CMA, including FY-4A AGRI: 14 channels, FY-4B: 15 channels, FY-4B GHI: 7 channels (6 in VIS/NIR).

CMA generate 14 RGB products according to EUMETSAT reference documents: GEO color, Natural Colour, Cloud image, Fog, Convective cloud, Airmass, Ash and Dust, Day-Night microphysics, Snow RGB.

The following parameters are used in this generation: RGB channel selection, Cross-calibration process, to have the same calibration levels, image processing: L1/L2 quality, observations angle, absorption correction, image enhancement. There are no intra-China variants, all products are full disc. AGRI-specific recipes may be needed.

B Connell: How do meteorologists use the products? - Meteorologists use for example the convective cloud product together with weather radar. What about seasonal adjustments? - Sometimes done.

FCI cases in the Middle East region (I Al Abdulsalam)

Ibrahim Al Abdulsalam (Oman Met Service) presented. Dust products are very important (FCI Dust RGB). Dust sources are the farms. Challenges are the dust height and thickness to interpret from Dust RGB product: They trust the same color in the product to forecast dust emission.

The other important one is fog and low clouds, helping forecasts for road traffic, aviation.

GeoColour : disappearance of fog during transition from night to day is considered unsatisfactory, asking for explanation.

FCI vs SEVIRI: with cloud phase RGB, small features are now seen with high spatial and temporal resolution.

Fire detection: the Cloud Phase RGB is used for detection.

B Connell noted that U.S. forecasters use a combination of Day and Night Microphysics RGBs, not the 24h Microphysics RGB.

Operational FCI RGBs over South Africa (Kanyisa Makubalo)

She presented on the operational RGBs at SAWS, based on EUMETView.

The GeoColour RGB is used in day and night shifts by the forecasters, even small-scale clouds seen in the day light. The True Colour RGB is used for convective cloud detection. Cloud phase RGB used for low level clouds, also shadows used to track thunderstorms. Dust RGB: highlighting the high-level clouds to distinguish from low level clouds.

Zoomed-in resolutions are not working well and making some pixeling (probably due to EUMETView). At times, animations are slowly rendering, possibly due to SAWS computing facility.

More education is recommended, for example more information on the colour scheme breakdown for each RGB directly in EUMETView. More awareness in the quick guides.

She pointed out that it is still very useful to have MSG-based RGBs available besides MTG.

Example of GeoColour use cases where MTG RGB products are missing some clouds visible in MSG RGB products.

MTG GeoColour used at nighttime, compared to MSG Fog/Low Cloud RGB. It is unclear why the Night Microphysics RGB is not used (available via EUMETCast Africa)? She made several recommendations to investigate the use of GeoColour visualisation.

G Holl remarked that there is the possibility to enhance GeoColour visualisation to avoid some issues presented.

V Nietosvaara asked about the main operational visualisation system at SAWS? - the SUMO tool is used for MSG. For MTG only EUMETView is used at the moment.

S Bojinski: GeoColour is not designed for night-time low cloud detection. You should have other RGBs for night-time detection via EUMETCast Africa service dissemination, with direct broadcast and terrestrial. EUMETView is not a tool meant for operational forecasting.

I Smiljanic noted the need for more user feedback from users on use cases of GeoColour RGB.

First experiences with the FCI Dust and Cloud Phase RGB (Jochen Kerkmann)

Jochen Kerkmann presented on FCI Dust RGB: Which AHI split window to be used for the dust RGB green component? The brightness temperature difference of (IR10.4 – IR8.6) of cloud water and ice is not as large as for the (IR11.2-IR8.6) channel difference. The same with ABI. Hence FCI is not as good as ABI and AHI in detecting cloud phase with Dust RGB.

Other dust RGB application: thin cirrus very well detected, so are moisture boundaries.

He investigated moisture detection from FCI VIS0.8-0.9 channels, and Cloud Phase RGB with NIR1.6, 2.3, VIS0.4, with several cases.

FCI High-resolution imagery at 500 m resolution is good for fog case in Central Europe.

G Holl asked for guidance how to handle different FCI channel resolutions within the same beam of an RGB.

Recommendation: in case two FCI channels are to be combined in the same colour beam of an RGB where one of them is available as High-Resolution imagery, the equivalent of that channel at Normal Resolution should be used to be compatible with the other channel resolution (i.e., avoid downsampling or upsampling).

Getting accustomed to the new FCI RGBs (Roxane Désiré)

Roxane Désiré presented on Meteo-France work in introducing seven new RGB products to forecasters. Many RGB products already exist but are used little. Meteo-.France has internal quick guides for FCI RGB use.

MF forecasters consider GeoColour RGB as very useful, for media, briefings.

Monitoring dust plumes over the sea: True Colour RGB rather than Dust RGB since the latter not very sensitive with FCI over the sea (somewhat disappointing).

Night Microphysics RGB from FCI applying recipe from Miguel-Angel Martinez (AEMET) – enthusiastic response by forecasters.

Cloud Type RGB – appearance of snow: yellow in dry conditions, reddish or greenish in moist conditions (if cloud-free).

Supercooled water cloud example as a major hazard for aviation presented. Cloud Type RGB was investigated to detect supercooled water clouds. Supercooled water cloud colour coding defined for RGB cloud type product.

B Connell informed that US forecasters were impressed by the resolution improvement, is this the case in MF also? - Yes, temporal and spatial resolution was much appreciated by MF meteorologists. We should remember the importance of the texture.

I Smiljanic picked up on the poor FCI Dust RGB performance over the sea: maybe better to use True Color RGB, because improvement there is huge above water bodies.

J Kerkmann noted that supercooled water cloud detection at high altitudes is best done with the Convection RGB due to 3.9-10.5 contrast.

JB Hernandez said that MF have removed the daytime use of the Night Microphysics RGB, and replaced it with True/GeoColour RGB during the day. Should the terms "Day/Night" be added to all RGBs in the naming convention?

RGBs in a testbeds for US weather offices (Christopher Smith)

Chris Smith (CISESS, U Maryland): GOES-R satellite liaisson. NWS and WPC using GOES and LEO for forecasting at Ocean Prediction Center. Most popular RGBs: AirMass RGB, Day Snow Fog RGB popular/useful at high latitudes, Nighttime Microphysics RGB used both at NWS and WPC, GeoColor popular for both centers (fresh snow) , ProxyVis useful for warm sea SST to detect low clouds and surface analyses, Experimental: Sea spray RGB (VIIRS), CrIS imagery RGB, OPC issues heavy freezing spray warnings. Sea spray is a major hazard for ship vessels (freezing). CrIS imagery RGB used by OPC for their SFC analysis. LxCast overlaid on the Day Cloud Phase Distinction RGB and GLM with ML helped to detect convective initiation in New Mexico 2-3 hours.

As RGB needs, he mentioned low level moisture gradients, GeoRing.

J Kerkmann wondered about CIRA loops in the slider that cannot be saved. Could we have a download option in MP? B Connell responded that there is a feature to download all images in one click and then you can do MP4.

B Connell: Forecasters mostly use split window for low level moisture detection. Do they look at dust product? - Forecasters do not look at the Dust product for that.

RGBs for monitoring of mid-latitude weather systems (Luciano Vidal)

Luciano Vidal is part of remote sensing group that works on R2O products and development training activities in collaboration of CoE Argentina and WMO RTC. 2017 Environmental satellite processing distribution system (ESPDS) but in place. SNM introduced imagery and RGB products based on GOES ABI in 2017.

Training program for the new generation GEO satellites with two stages (virtual and face-to-face).

RGBs: RealColor, GeoColor, NaturalColor, Airmass, Day Convection, Clouds, Night MP, Ash. Recipes based on EUEMTSAT and CIRA. (Which recipes are exactly used, and for which imager?).

Typical mid-latitude weather systems over Argentina are: Extra tropical cyclone, MSC, supercell, Fog and low-level stratus, Dust storm, Ash, severe weather.

In MCSs, sometimes convection reaches 20 km height. Also, supercells are associated with hail and sometimes with tornadoes. Shallow convection observed during austral winter with graupel and snow showers. Water vapor channel and airmass channel used.

GeoColour RGB "Geo2Grid" based on SSEC. With this product, dust plumes are also detected. Not operational yet. Important for visibility on highways.

Natural color RGB detecting snow in Patagonia. Ash RGB is very important for VAAC Buenos Aires.

SNM are mixing with EUMETSAT and CIRA recipes for ABI and FCI data. They started using all EUMETSAT recipes. CIRA consulted them and then they modified some of the parameters. Thus, it is a mixture and confusing some forecasters.

I Smiljanic asked about the difference between Real color RGB, GEO color RGB - these are made for media coverage and public use.

J Kerkmann: Many cases studied in Argentina because there is lot of convection. And it would be useful to look at both satellites to take the benefit of slant view of MTG vs GOES. - Indeed Meteosat is used a lot at SNM.

The mix of CIRA and EUM recipes in use is a result of evolution of service: first started with EUM, then evolved as CIRA recipes became available. At first implementation, they were not clear about the standards (or lack thereof).

RGB products in high-impact cases (Regina Ito, Diego Souza)

Regina Ito has a seasonal forecasting background and has been working on satellite applications and RGB composites for the past 10 years. Diego Souza heads the VLab CoE in Brazil and showed the reception and visualisation system DSAT (https://www.cptec.inpe.br/dsat) only based on open-source tools. They presented this work in the WMO data processing and visualisation task force for RA III/IV. A range of natural disasters occur in Brazil. She highlighted the major floods in Rio Grande do Sul in April/May 2024, with 2-week accumulated rain amounts of up to 50% annual climatology (500-800mm). According to NWP data moisture flux enabled by strong atmospheric rivers. She showed overlay of Night Microphysics RGB, Day Cloud Distinction RGB, and wind fields. True color RGB was used to monitor the movement of the sediment from inland to the Ocean. INPE is part of Internatinal Chater Space and Major Disasters, which was invoked and several information products were produced to inform civil protection and recovery agencies.

Detection of fires and smoke with True Colour RGB and Fire Temperature RGB. More than 2500 fire hot spots were detected in one month.

Ash RGB with IR 10.5 and 11.2 was tested. For Night Microphysics RGB they use seasonal tuning between summer and winter.

They see issues related to training with missing basic concept of channel properties and basic concept of RGB technique, and lack of practical skill of RGB interpretation.

Day and night RGBs are preprocessed as transparent for DSAT that any of them can be overlaid.

Action: Ivan to check if FCI data is available for May 15 2024 case study.

NOVEL RGB DEVELOPMENTS AND RGB ENHANCEMENTS

Plans for Himawari-10, novel RGB composites (Akihiro Shimizu)

Shimizu Akihiro introduced Himawari-10 to be launched in 2028. Himawari-9/AHI to operate until 2029. Examination of RGB composites that can be generated with GHMI/Himawari-10. He reported on investigation of new bands on GHMI:1.38 and 5.1 um.

1.38um band used with GHMI has been used with MODIS and VIIRS imagery.

Investigation of Cloud Phase RGB, and Arctic Day Cloud RGB (sensitive to cloud phase).

S Bojinski commented on the GHMS/Himawari-10 sounder, and on the importance of RGB guides and recipes. Can the two variants for example ash, dust, 24microphysics RGBs for AHI be reduced to one? Which variants you would recommend? - 11.2um channel is more effective to use in RGB composite and should be used.

B Connell remarked that the choice of band may be dependent on the chemical composition of dust.

Cloud Analysis using Meteorological Satellite Data (Shiro Omori)

Shiro Omori (JMA) presented Himawari AHI imager with 16 channels. HimawariCast has 34 user countries via commercial communication satellite, 24 HimawariCloud users, and a web service covering 42 countries and territories.

RGB quick guides have been published on JMA website including WMO recommended RGBs, LEO satellite RGS and JMA specific RGBs.

Training provided in AOMSUC and as bilateral training courses for NMHSs which have HimawariCast installed.

Book published on Utilization of Satellite Imagery by JMA, including section on RGB composite imagery. It can be downloaded online on JMA's website. Details on the RGB recipes are provided not in the book but in the JMA RGB quick guides: Meteorological Satellites -Japan Meteorological Agency (JMA).

B Connell asked whether the book was included on the WMO ETR website or linked to WMO satellite skills document.

Niche RGBs developed at CIRA and at NOAA (Curtis Seaman)

Curtis Seaman expanded on the subjectivity of RGB and colour schemes, depending on eyesight, display devices, printer quality. Not all are sensitive to specific colors and also displays have variability to visualize the color details.

GeoColour RGB city lights layer, low clouds mask (night), high clouds mask. "Gained" one hour by improving representation of the terminator. GeoProxy blends ProxyVis with GeoColour RGB; ProxyVis is a multiple-linear regression model to blend IR bands (3.9 µm, 10.4 µm and 12.2 µm) to replicate VIIRS Day/Night Band images.

Fire Temperature RGB is designed to represent active fire intensity in an intuitive manner; recommendation increasing the red max to 343 K.

- VIIRS version based on radiances (rather than BTs or reflectances)
- AWIPS version,

The Day Fire RGB combines VIS, 0.8 µm band and 3.9 µm band. Recommendation to use 3.9 um reflectance value, for better contrast between active fires and background. AWIPS cannot handle that so 273-343 K, gamma=0.4 used. Alaska Fire Service use this at very high resolution based on VIIRS, more sensitivity to smoke.

NGFS microphysics RGB is tuned for detecting low-intensity, smouldering fires,for early detection of fire initiation.

- Snowmelt RGB uses 1.24 um band on VIIRS/METImage for freezing rain and sleet.
- Blowing Snow RGB, as modification of the Day/Night
- Sea Spray RGB

J Kerkmann recommended to put the 0.4 channel on Blue beam, rather than 0.6? - It uses 0.6 because of the resolution of the channel.

G Holl asked whether images cannot be preprocessed and then displayed in AWIPS? - Because of bandwidth issues and NWS not being able to accept large amounts of imagery from outside.

Which areas of your instruments/RGBs are covering, over Antarctica?- All products are available for polar orbiters and over Antarctica.

Three variations of fire RGB products to cover fire intensity, details, burned areas and smoke.

Demonstrating the power of the TPW RGB (HP Roesli)

HP Roesli: He presented SMT (smt = solar moisture transmission) ERA5 850&925 hPA analyses introducing blue-green-yellow color scale. The capability of the 0.9um channel to total moisture variabiltiy is demonstrated for a dry line over the Baltic Sea.

“TPW” RGB example for cirrus cloud intrusion from Atlantic; could nicely see effect of moist air on snow appearance in Cloud Phase RGB from yellow to green.

Strong recommendation to use ration 0.9/0.8, instead of subtracting the channels.

S Bojinski remarked that the ratio discussed with ESSL colleagues and related terminology, and it should be called “Total Moisture Imagery” or “Water Vapour Transmittance” to make clear that it is not a quantitative product. Whether an RGB should be recommended is for the discussion, as is how to do cloud masking.

Novel findings for FCI RGBs (Ivan Smiljanic)

FCI sees three oceans, extending even to the Pacific coast of Chile in South America. 3D impression of clouds in FCI True Colour RGB in high latitudes.

High realism of Cloud Phase RGB for convection captured by FCI proven by S-2 ground truth.

Shadows in IR imagery can be useful.

New RGB products development or RGB tuning (Jean-Baptiste Hernandez)

JB Hernandez: He explained MF product development process from EUMETSAT raw data to product dissemination to users using FCI and SatPy together with algorithms (including RGB recipes). Comparison and evaluation of the products is based on web tools (Jupyter notebook) to compare products with different recipes. The tolls were developed by Rudy Coste. For the operational use, also the data processing time is crucial.

GeoColor RGB fine tuning for day and night-time. better day-time transition, by night: no blue clouds, more opacity on thigh clouds.

Snow term in GEOColor example.

Jupyternotebooks for comparison of RGBs: 2 or 4 RGBs.

“General RGB” is a colour matrix, as a basis for creating RGBs: Night part of GeoColour RGB, day part of GeoColour RGB.

FireTemp RGB + GEOColor: Difficulty to isolate the fire signal from the Fire Temperature RGB, to add a layer to the GeoColour RGB. City lights get in the way as well, and they have been removed.

GEOColor planisphere with 500 m spatial resolution.

Snow mask, is that a static layer? - It is updated every 10 min.

New FCI, synthetic NWP RTTOV and Future IRS RGBs (Miguel-Angel Martinez)

Miguel-Angel Martinez covered:

A training and validation dataset of collocated real and synthetic clear-air brightness temperatures from MTG-I/FCI and MSG/SEVIRI (July 2024 - January 2025) was generated using the NWCAF iSHAI 2025 environment and its PGE00 tool. This dataset was used by Miguel-Angel Martinez to tune SEVIRI recipes for FCI IR channels, as highlighted in M. Putsay's presentation and proposed as the tuned FCI RGB recipes of this workshop.

The Night Microphysics RGB (Fog) case demonstrates the tuning process. After regressing SEVIRI to FCI and applying this to SEVIRI RGB thresholds, new adjusted FCI recipe thresholds were obtained. Comparing FCI RGB fog images from both recipes showed improved color separation between low clouds/fog and higher clouds. An animation of high-resolution FCI channels with the tuned FCI fog recipe is also presented.

Then he showed how synthetic BTs with 3D transmittance profiles and weighting functions aid RGB training. For instance, in dry atmospheres, NIR1.3 channel signals from the ground can cause reddish tones in Cloud Type RGB. This also helps compare similar MSG and FCI channels, revealing FCI WV6.3 penetrates deeper than SEVIRI WV6.2.

Early results are presented on correcting Air Mass RGB dependence on satellite zenith angle. This involves using simulated FCI synthetic BTs (0°-75° zenith) every degree (made on calculation of NWCSAF iSHAI new coefficients for FCI) to regress RGB thresholds against 45° zenith angles. Applying these thresholds by zenith degree eliminates characteristic purple tones at high zenith angles in corrected Air Mass RGB images. This correction will be generalized for 76°-90° and extended to other geostationary instruments (GOES/ABI, Himawari/AHI) to cover the entire GEO ring. Generalizations to other RGBs will also be studied.

Future activities include generating new RGBs when hyperspectral MTG-S/IRS real BTs become available. Examples using IASI proxies and synthetic RGBs are shown. One important example, based on an idea by Justin Sieglaff and Paul Menzel, is using spectral lines that change from absorption to emission during temperature inversions. They called these lines the "emperor's finger": thumbs up for emission (inversion) or thumbs down for absorption.

Long-term, new RGB generation methods are being explored. This includes using nonlinear functions with a variable number of channels (e.g., neural network outputs) for R, G, and B components, given good training datasets (like using IRS L2) to be generated. Another further possibility is training RGBs by specifying desired outputs in training sets.

These efforts will require an environment to evaluate millions of potential RGBs from the new IRS instrument to select the most advantageous ones.

Correlation of TPW field with FCI 0.9/0.8 ratio is analysed.

M Raspaud: Problem with Airmass RGB at limb in Sweden. You do normalization?

He answered that normalization of correction is made in the sense the color on airmass RGB are similar over Sweden than in Iberian Peninsula and they are not shown with purples tones.

S Bojinski: Proposed recipes. Is it safe to round numbers in RGB range? - Can be rounded up to one digit after the comma.

Corrections and enhancements in True/GeoColour RGB (Johan Strandgren)

Johan Strandgren presented development of True Colour RGBs from FCI implemented in SatPy, and used in EUMETView. The correction steps are:

Step1: Raw RGB
Step2: Log scaling
Step3: Sun zenith angle
Step4: Rayleigh correction

- FCI 0.51 um (green) does not capture the local reflectance maximum of chlorophyl. The solution is a NDVI-weighted hybrid green correction method. It is based on previous work for AHI.
- Testing green correction with NDVI hybrid green with OLCI data as a reference.

Step5: NDVI hybrid green correction

FCI GeoColor: for Daytime TrueColor RGB, for nighttime NASA Marble as static background.

He mentioned known limitations and issues of GeoColor listed (see presentation). Next steps: reduce brightness of black marble background, tuning of sun zenith angle correction, improve night-time low-level cloud and fog detection. FCI commissioning still limits the resources for this work.

S Bojinski: Can we say that the use case for GeoColour is broader than media and public relations? - You need several RGBs for different applications. With single RGB you always miss some details. It is true that it is not only for public relations use.

M Raspaud: Good that you showed all correction steps, which applies also for other RGBs.

C Seaman: U.S: NWS is also using NDVI correction approach.

J Kerkmann proposed that EUMETSAT should consider a blended product including day microphysics and night microphysics. (24h microphysics is already a blended product).

H Roesli: A distinction should be made between professional RGB products, and other RGB products for public relation use.

Early visibility of GeoColour, for example on EUMETView, has triggered its use in several contexts including forecasters, and SAWS for example. Although the main use case is media and public outreach, feedback suggests that users would like to use it for fog detection, for example.
The use case for each standard RGB should be made very clear!

G Holl: users of EUMETView should have easier/direct access to RGB quick guides.

MAJOR CHALLENGES PER EACH RGB

Name limitations or "This RGB also detects _X_ and it is very useful" (Bernie Connell)

Bernie Connell described the SO2 RGB and its sensitivity to SO2 at high or low levels. She proposed to rename it to "Volcanic Emissions & Dust RGB" because other features are also visible. The group agreed to the name "Volcanic Emissions (SO2) RGB".

BREAK-OUT GROUPS

Before breaking into groups, Zsofia Kocsis presented the EUMETrain guide on generating RGB composites: How to create or adapt an RGB scheme that provides general guidance.

Starting in plenary, some clarifications about the document: - the table with Full and Shorthand names was introduced. Plenary discussion on the AHI Airmass recipe served as an exercise.

Notes on AHI Airmass from plenary:

- J. Kerkmann. Remove "inverted" and make clear in the definitions that range can be inverted, i.e., remove the terms "MIN" and "MAX". Airmass special version for tropics to avoid saturation. Otherwise OK.
- HP: what level data is the starting point? 1.5?
- Bernie: In Americas level 2 data used (BT).
- J. Kerkmann: keep RGBs intact and original, caution with corrections
- G. Holl: would this then apply also Rayleigh correction?
- JK: solar zenith angle correction is necessary in loops with morning and midday.

General Recipe (for inclusion in Guidance document – Annex III):

- Input data are the default calibrated radiances measured by spectral imagers provided by each satellite operator in near-real time.
- Convert counts into radiances (formula provided by each satellite data provider)
- Then convert these into BT (IR) using Planck function
- No limb cooling correction applied
- SZA correction needed for loops, to be applied to solar channels, using the method proposed by Li & Shibata (2006) – see presentation by J Strandgren

The group split up into three groups, addressing:

- Basic RGBs: Martin Raspaud, Jochen Kerkmann, Roxane Désiré, Jian Lu, Marianne Patzer, Zsofia Kocsis, Stephan Bojinski
- Special Applications RGBs: Kanyisa Makubalo, Akihiro Shimizu, Shiro Ohmori, Bernie Connell, HP Rösli, Diego Souza, Regina Ito, Rudy Coste, Johan Strandgren, Curtis Seaman, Mária Putsay

- RGB Variations and other RGBs: Gerrit Holl, Ivan Smiljanic, Vesa Nietosvaara, Marielle Bjerring Fournier, Chris Smith, Jean-Baptiste Hernandez.

Classification of RGBs for the purpose of this workshop (cf Annex III):

- Basic RGBs: applicable everywhere to weather forecasting (globally), recommended as core part of operational production, display systems and forecaster training
- Special Applications RGBs: applicable to specific contexts and applications (fires, dust, ash, SO2, snow), recommended for operational production, display and training on an on-demand basis
- RGBs under development: require further work and discussion, not recommended for operational production, display and training
- Legacy RGBs: RGBs that are still used operationally but have been superseded by updated or better RGBs, and that are therefore not recommended for further development
- Other visualisations: common visualisations of satellite data that are not RGBs

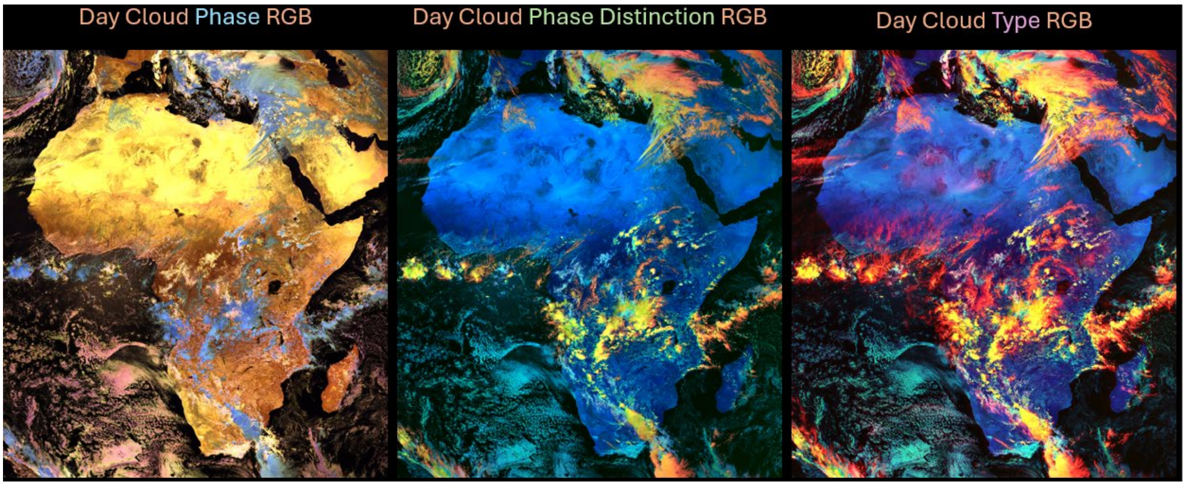
REPORTING TO PLENARY

The groups subsequently reported to Plenary to get their recommendations agreed. Results are available in Annex III.

Discussion on Cloud Phase, Cloud Type and Cloud Phase Distinction RGBs

We discussed the naming of the Day Cloud Phase, Day Cloud Phase Distinction and Day Cloud Type RGBs. The Day Cloud Phase Distinction RGB is very similar to Day Cloud Type in terms of RGB purpose and colours. The only difference is that the Day Cloud Phase Distinction RGBs uses the IR10.3 channel for the red beam instead of the NIR1.3 channel for Cloud Type. This is because the NIR1.3 channel was not available when the RGB was developed (for AHI). Given the similarities with the Day Cloud Type RGB, and clear differences compared to Day Cloud Phase RGB, the group agreed to put the Day Cloud Phase Distinction RGB as a variant of Cloud Type RGB and in the documentation say that if the IR10.X is used for the red beam, the RGB is also known as Day Cloud Phase Distinction RGB. It was noted that in the U.S., the IR10.3 channel is used for the red beam even though the NIR1.3 channel is available on ABI. Some users might be reluctant to change and use the NIR1.3 channel instead and/or change the name to Day Cloud Type RGB. This requires further discussion.

The figure below shows a comparison of the three RGBs to highlight the similarities of the Day Cloud Phase Distinction and Day Cloud Type RGBs, and the added value of using the NIR1.3 channel instead of IR10.X for thin cirrus.



RGBs under development

Regarding the presentation by HP Roesli on the "smt RGB": the Total Moisture Imagery based on the 0.9/0.8um channel ratio using the beige-green-blue colour palette has been implemented by ESSL in their Weather Displayer, with good feedback by forecasters. However, it needs further validation, and it is not an RGB or quantitative product.

J Kerkmann proposed an RGB composite for development using the 1.3um channel. The group noted that this channel is affected by Earth stray light for FCI on Meteosat-12.

C Seaman mentioned a dynamic RGB using polar-orbiting Day-Night band imagery to detect power outages.

Other such RGBs are listed in Annex III.

RGB applications

M Raspaud suggested to add a table mapping application areas against relevant RGB composites: Meteo-France have suggested such a mapping, and a similar exercise was undertaken at the 2022 workshop (refer to 2022 workshop report to map topics against RGB standards).

Action: I Smiljanic to poll participants online regarding their opinion on application areas vs RGB standards.

M Raspaud pointed out the need for transparency when publishing the "living document" that includes all outcomes from the workshop, including calls for further investigation and development (WMO should be owning this document).

In addition, a WMO Guideline shall be published using the Basic and Special Application RGBs (only). For completeness and better guidance of users, the WMO Guideline shall contain an Annex listing "Common Visualisations of Satellite Data that are not RGBs".

The meeting report will recognise common visualisations that are not RGBs: GeoColour (Annex III).

Solar zenith angle correction (Johan Strandgren)

J Strandgren showed the effects of simple (1/cos(sza)) and Li & Shibata (2006) solar zenith path length corrections on imagery near the limb.

See presentation for the proposed text of the solar channel normalization recommendation. The corrections will be further tested with FCI data.

After investigating the correction effects and gain of information at the limb edge from both methods, the group recommended the Li & Shibata (2006) correction method as standard:
correction_factor = 24.35/(2·COS(SZA)+ SQRT(498.5225· COS(SZA)^2+1)

IMPLEMENTING THE RGB STANDARDS

Implementation of (standardized) RGBs

Martin Raspaud explained the motivation for an unambiguous description of recipes.. The SatPy composite configuration was presented. You can include description, references in the configuration file. The Satpy enhancements part includes modifiers (corrections, channels, std atmosphere).

Sentinel 1 SAR sea ice example was presented, creating RGBs with two channels (H and V).

Configuration system:

Pros
- Yaml file easy to read with any language
- Extendable
- Metadata included
- Can work with wavelengths

Cons
- Relies on satpy Python functions
- Separation between composite and enhancement parts is confusing

Power of configuration was demonstrated with an example.

Proposal for sharing the RGB standards:
- Machine readable version to be created
- Implementation agnostic
- Relying on metadata
- Version controlled
- Ideally, can create a human-readable version

JSON commonly used for serialize data/web api outputs, which has extension for Maths.

Known unknowns:
- Input data need to be agreed
    - To be specified in JSON file
    - Reflectance and details to be defined in the document
- How we describe algorithms/correction
    - Links to reference documents
- Standard names for
    - Platforms
    - Sensor
    - Channel names
    - Common quantities (e.g. Sun Zenith Angle)
- Who hosts the human-readable recipes?
- Hosting
    - Version control
    - Public, everyone able to propose changes
- Maintenance and governance
    - Who maintains it?
    - Who decides proposed changes?
    - When to release new versions?

Bernie: “Channel” vs “Band” to be used? In the document we use channel; the two terms are equivalent, and distinct from “Wavelength” which is a physical quantity independent of technology (cf. the use of “VIS0.8”, “VIS0.9” FCI channels since the detector and focal planes are called “VIS” for these channels).

What should be a level of maturity of RGB to be approved? M Raspaud: Used by community first of all together with agency prework.

J Strandgren: Corrections are difficult to handle but user should have possibility to use or not.

M Raspaud: the Github platform to be used also for the WMO Guidance document itself.

JP: Hardware requirements for Satpy? M Raspaud: Virtual machine with 4 cores, 128 GB used for European weather cloud. Two minutes for a composite FCI data processing

G Holl: is there an audience for JSON format? - There are request for this.

I Smiljanic: Will support it but it is quite ambitious and needs resources.

S Bojinski: EUMETSAT might be able to support the effort to set this up (e.g., training placement).

J Strandgren: How to keep document and machine-readable format aligned?

Pre-workshop Questions:

GEO-ring RGB development:

- AGRI has been added
- Others to be added subsequently

Action: Check with Jian Lu (CMA) whether AGRI-specific recipes should be added where they are missing in Annex III.

Parallax correction made for RGBs?

- Meteo-France are doing it, already for SEVIRI and continued with FCI: it is a challenging topic they are not happy yet with the results
- Parallax correction code was shared with G Holl, DWD does not use it at the moment
- Agreed not to include as recommendation for RGB since it creates lot of issues

Promotion, Awareness

B Connell recommended to share the workshop report in wider relevant networks.

K Mukabalo suggested to have more operational experts participating in RGB workshops.

Quick guides

M Putsay presented the EUMeTrain resources: Recently, FCI quick guides were created for the new channels and basic RGBs. Quick guides do not include recipes, but there is a link to another document including the recipes. The MTG RGB training part is under development. G Holl noted that current examples are for VIIRS and SEVIRI. Is there a plan to extend to cover FCI? - Yes, at least some of the guides will cover FCI in the future.

Extended guides

EUMeTrain: Extended guide to Cloud Phase and Cloud Type RGBs were presented by M Putsay. She is also working on an extended guide for Fire RGB.

RGB tool

EUMeTrain tool (see below).

DISPLAY SYSTEMS AND TOOLS

Pytroll/Satpy (Martin Raspaud)

M Raspaud presented the collaboration with other Nordic countries and Baltic States (NORDSAT, 8 countries) in operational processing and visualization of RGBs. Using the same software for preprocessing, open source, Pytroll. These tools are taken into European Weather Cloud EWC. Data is served into WMS and Mapserver. Everything they do they provide in containers. Github repo in use. Accessing the data through Open GeoWeb. Day/night where night part of the data made transparent for compositing with other data sets. Tool supports also split windows.

G Holl: Is it cloud optimizes when it is so fast. - Yes, it uses cloud optimized Geotiffs.

V Nietosvaara: Does it support WMS? - Yes, it supports.

EUMETRAIN (Maria Putsay)

EUMETRAIN RGB tool can be downloaded from the website. You can use it to analyze RGB image related to specific RGB and instrument (One can retrieve the calibrated values). With the parameter file you can save the recipe parameters of the RGB.

Meteo France tools (Rudy Coste)

They have internal tools to support the RGB development to be used by the forecasters. The tool supports magnifier feature to compare the visualization outcome of the different channels in one RGB. There is also a mouse tool to compare the visualization of the several RGB products in one geographic location. There is also a slider tool to compare for example different RGB correction algorithm visualizations. MF is also testing parallax correction algorithm and tool is used to visualize the effect of the correction algorithm.

The application is available at EWC.

http://jupyterhub.fr-mf-general.s.ewcloud.host/index.html (not secure site, may not work on all browsers)

SIFT (Johan Strandgren)

J Strandgren reported that there is ongoing development at EUMETSAT. New features are added for improved user experience. A short course for SIFT tool is available on EUMETSAT website: https://classroom.eumetsat.int/course/view.php?id=478. A local version is available to download.

DSAT (Diego Sousa) https://www.cptec.inpe.br/dsat/

He presented DSAT tool developed by INPE. Pan/zoom and overlay features were presented with RGB examples. There is a slider function and “magnifier” to compare two different RGBs. You can also overlay alerts by weather service and NWP fields like winds. WMO RAIII/IV group website include support material how other institutes can develop the same localised visualization.

For supporting training and developers they use GoogleCoLab using Python. JupyterLab was also recently installed supporting future training needs.

M Raspaud: Is there a limitation of the number of notebooks? - Limitation is the google account disc space.

Meteo France training material (Roxane Désiré)

Roxane presented their internal chat channel and communication tools for collecting feedback on RGBs in use. They also have 12 Quick Guides available. RGB product names are kept in English even if other content is in French in the guide documents. It gives guidance how the product is used, visualization examples, limitations, comparison with other products etc. Guides are focusing on MTG data only.

Meteo-France has user guides for all their MTG products, including NWC SAF products openly available.

Meteo-France also publish monthly newsletter containing anomalies detected, instrument limitations and effect on RGB products.

Meteo-France has internal training the trainers sessions to anchor the use of MTG in the forecaster's work. There are 400 forecasters at Meteo-France.

CIRA Slider (Curtis Seaman)

Curtis presented CIRA Slider features. He notified that GOES-16 will be shut down and replaced with GOES-19. CIRA Slider data will accommodate with that change. There is a polar tool for JPSS on CONUS. You can use that for the comparisons between JPSS and GOES data in the Northern hemisphere sector. CrIS imagery products were added recently. There is a feature to download all images in zip file for your animation. You can contact Curtis if you want additional products to be added. FCI products are still missing.

DMI: Can you see only one orbit at time? - There was no request for that so far but visualizing always the most recent per location.

Next meeting

Proposal to have the next meeting in late 2027 / early 2028. Meteo-France proposes to host.

It was further suggested to have an annual online catch-up meeting.

ANNEX I: LIST OF PARTICIPANTS

Regina Ito, INPE
Diego Souza, INPE
Luciano Vidal, SMN Argentina
Bernie Connell, CIRA Ft Collins, CO, USA
Maria Putsay, EUMETrain, HungaroMet (retired)
Curtis Seaman, CIRA Ft Collins, CO, USA
Christopher Smith, CISESS, University of Maryland, USA
Jian Lu, CMA China
Roxane Désiré, MeteoFrance
Rudy Coste, MeteoFrance
Jean-Baptiste Hernandez, MeteoFrance
Marianne Patzer, DMI
Marielle Fournier Henckel, DMI
George Murin, SMHI
Marie Staerk, SMHI
Ibrahim Al Abdulsalam, Met Service Oman
Zsofia Kocsis, EUMETrain, HungaroMet
Johan Strandgren, EUMETSAT
Ivan Smiljanic, EUMETSAT
HP Rösli, Expert, retired
Jochen Kerkmann, Expert, retired
Vesa Nietosvaara, EUMETSAT
Heikki Pohjola, WMO
Stephan Bojinski, EUMETSAT
Gerrit Holl, DWD
Hampus Sellman, SMHI (online)
Miguel-Angel Martinez, AEMET (online)
Rosen Penchev, Bulatsa, Bulgaria (online)
Akihiro Shimizu, JMA Japan (online)
Shiro Ohmori, JMA Japan (online)
Kanyisa Makubalo, SAWS, South Africa (online)
Corine KarroBarro, Swedish Armed Forces (online)

ANNEX II: AGENDA

RGB Workshop co-hosted by SMHI, EUMETSAT and WMO
1-3 April 2025
SMHI, Norrköping, Sweden

Workshop objectives:

Consolidate contemporary GEO-ring RGB composite standards and publish them in a workshop Report

- Identify novel RGB products
- Demonstrate applications RGBs using new-generation imagers (eg FCI)
- #Identification of common platforms to share the RGB recipes and related material

Day 1:

Time	Discussion items	Contributor
09:00-09:15	Welcoming Remarks and Logistics	SMHI
09:15-09:45	Purpose and Scope of Meeting: Setting the scene	Stephan Bojinski, Heikki Pohjola
09:45-10:15	Overview of RGB composite recipes, applications, limitations	Maria Putsay, Ivan Smiljanic
	15' Coffee break	
10:30-12:30	New application findings for well-established RGBs (10'+5'):	
	Regional use of Himawari RGBs	Bodo Zeschke
	RGB applications at CMA	Jian Lu
	FCI cases in the Middle East region	Ibrahim Al Abdul Salam
	Operational FCI RGBs over South Africa	Kanyisa Makubalo
	First experiences with the FCI Dust and Cloud Phase RGB	Jochen Kerkmann
	FCI Dust RGB perspectives	Djordje Gencic
	Getting accustomed to the new FCI RGBs	MeteoFrance
	RGBs in a testbeds for US weather offices	Javier Villegas
	RGBs for monitoring of mid-latitude weather systems	Luciano Vidal
	RGB products in high-impact cases	Regina Ito
12:30-13:30	Lunch break	
13:30-15:30	Novel RGB developments and RGB enhancements (10'+5'):	
	Niche RGBs developed at CIRA and at NOAA	Curtis Seaman
	RGBs constructed from observations and products	Andrew Heidinger
	Demonstrating the power of the TPW RGB	HansPeter Roesli
	Different concepts for FCI solar RGBs	Ivan Smiljanic
	New RGB products development or RGB tuning	MeteoFrance
	New FCI, synthetic NWP RTTOV and Future IRS RGBs.	Miguel-Angel Martinez
	Plans for Himawari-10, novel RGB composites	Akihiro Shimizu
	15' Coffee break	
15:45-17:00	Major challenges per each RGB – Intro	Ivan Smiljanic, Maria Putsay

	Name limitations or 'This RGB also detects _X_ and it is very useful'	Bernie Connell

Day 2:

Time	Discussion items	Contributor
09:00-09:15	Instructions and break-out room division	Facilitators
10:15-11:15	(Expert presentations +) Group discussions: RGB standardisation per/across sensor/s: recipes, names and enhancements	All
	15' Coffee break	
11:30-12:30	Reporting out to Plenary	Rapporteurs
12:30-13:30	Lunch break	
13:30-15:45	(Expert presentations +) Group discussions: RGB standardisation per/across sensor/s: recipes, names and enhancements	All
	15' Coffee break	
15:45-17:00	Reporting out to Plenary	Rapporteurs

Day 3:

Time	Discussion items	Contributor
09:00-09:15	Instructions and break-out room division	Facilitators
09:15-11:15	Group discussions: RGB standardisation per/across sensor/s: recipes, names and enhancements	All
	15' Coffee break	
11:30-12:30	Reporting out to Plenary	Rapporteurs
12:30-13:30	Lunch break	
13:30-14:00	Implementing the RGB standards	Martin Raspaud
14:00-15:00	Training repositories: Quick guides, Extended Guides, RGB tool, Jupyter Notebooks, Self-learning modules, Other guidance	All
	15' Coffee break	
15:00-16:00	Display systems and tools – news and updates: AWIPS, GEarth, RAMMS Slider, Pytroll/Satpy, SIFT, EUMETView, other	All
15:00-16:00	Draft Meeting Report and Recommendations	Heikki, Bernie

ANNEX III: RGBS AND OTHER VISUALISATIONS RECOGNISED AT RGB 2025 WORKSHOP

1. RGB RECIPES AND OTHER VISUALISATIONS RECOGNISED AT RGB 2025 WORKSHOP

Definitions

- **Basic RGBs**: applicable everywhere to weather forecasting (globally), recommended as core part of operational production, display systems and forecaster training
- **Special Applications RGBs**: applicable to specific contexts and applications (fires, dust, ash, SO2, snow), recommended for operational production, display and training on an on-demand basis
- **RGBs under development**: require further work and discussion, not recommended for operational production, display and training
- **Legacy RGBs**: RGBs that are still used operationally but have been superseded by updated or better RGBs, and that are therefore not recommended for further development
- **Other visualisations**: common visualisations of satellite data that are not RGBs

Basic RGBs
Airmass RGB
24h Microphysics Cloud RGB
Night Microphysics RGB
Day Cloud Phase RGB
Day Microphysics RGB
Day Cloud Type RGB

Special Applications RGBs
Day Severe Storm RGB
Overshooting Tops RGB
24-hour Microphysics Dust RGB
Simple Water Vapour RGB
24h Microphysics Ash RGB
Volcanic Emissions (SO2) RGB
True Colour RGB
Fire Temperature RGB
Day Fire RGB
Day Snowmelt RGB

RGBs under development
Day Moisture RGB
24-hour Fog Dynamics RGB
Day Blowing Snow RGB
Day Sea Spray RGB

Legacy RGBs
Day Land Cloud RGB
Day Microphysics RGB (with NIR1.6)
Day HRV Cloud RGB
Day HRV Fog RGB
Day Snow-Fog RGB
Differential Water Vapour RGB

Other visualisations
GeoColour Visualization

Recommended WMO standards for RGB composites/recipes from geostationary meteorological imagers

Contents

DEFINITIONS: RGB COMPOSITES, RECIPES AND GAMMA CORRECTION

This description explains how to create the WMO-recommended standard RGB composites from the new-generation imagers on meteorological satellites in geostationary orbit ("GEO ring") operated by the U.S., Europe, China and Japan that are most relevant for operational weather forecasting. An RGB composite here is a combination of spectral channel radiances or derived quantities (reflectance in %, brightness temperature in Kelvin degrees), measured by a passive multi-spectral satellite sensor, that are projected into the Red-Green-Blue (RGB) colour space and sometimes tuned using a gamma correction for enhanced display. The standards are based on community guidance developed over the past two decades (Ref. Workshop reports 2007, 2012, 2017, 2022, 2025). While the terms "channel" and "band" are equivalent, "channel" is used in the following.

The table below shows the parameters needed to define an RGB composite (a "recipe"). General guidance is provided, e.g., by the resource "How to create or adapt a RGB scheme" (EUMETrain).

Colour beam	Channel (or combination)	Range			Gamma
Red	...	Red_1	Red_2	K or %	Gamma1
Green	...	Green_1	Green_2	K or %	Gamma2
Blue	...	Blue_1	Blue_2	K or %	Gamma3

Template of the parameters used in RGB "recipes"

The second column shows which channels (or channel combinations) should be visualized in the red, green and blue colour beams.

- Input data should be the default calibrated radiances provided for each channel by the satellite operator in near-real time, to be converted into reflectance (%) or brightness temperature (BT) as per a formula provided by the satellite operator. In case digital counts are provided for each channel by the satellite operator in near-real time they need to be converted into radiances (formula provided by each satellite data provider)
- For solar channels, the calibration should include a solar zenith angle (SZA) correction, using the method proposed by Li and Shibata (2006)[1]. This method results in smoother imagery and preserves more information than a simple 1/cos(SZA) correction at the terminator.
- Correction for the effects of Rayleigh scattering can be done for channels with short wavelengths that are sensitive to atmospheric scattering (e.g. around 400-600 nm). This is recommended for generating the True Colour RGB, whereas for other RGBs, such a correction is considered optional, especially for the VIS0.6 channel since the effect is small and the correction is adding extra computational time.
- No limb cooling correction should be applied.
- No parallax correction should be applied.

[1] Li, J. and Shibata, K. (2006). On the Effective Solar Pathlength. *Journal of the Atmospheric Sciences* 63(4) pp. 1365-1373

Then the images should be *enhanced* using the Range, i.e., columns 3 and 4 of the above Table (e.g., Red_1, Red_2) of R or BT values to the full range of display values (0–255, BYTE) using a linear stretching and possibly a non-linear stretching (in case Gamma ≠ 1).

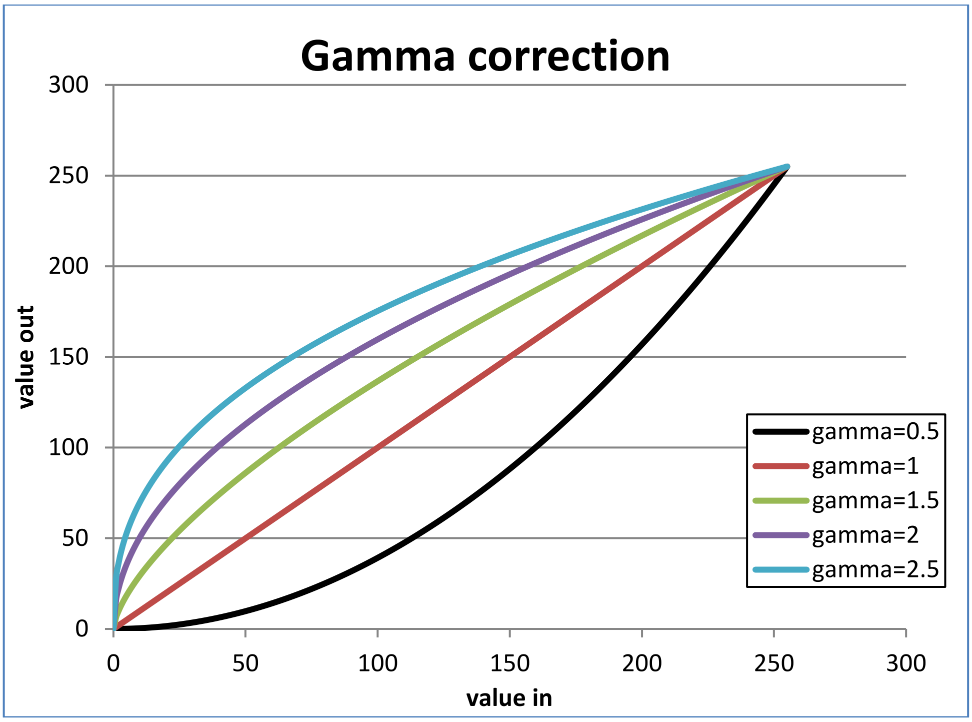
- The images should be *linearly stretched* within the brightness temperature or reflectivity ranges. (The 3rd and 4th columns of Table 1 contain the limits of the corresponding ranges, while the 5th column contains the unit. In most cases, the value in the 3rd column (e.g., Red_1) is smaller than the one in the 4th column (e.g., Red_2). In some cases, the range is 'inverted': the value in the 3rd column is larger than the one in the 4th column.)
- In some cases, *non-linear stretching* is also needed. In most such cases, a so-called gamma correction is performed. If the gamma parameter is greater than 1 then the image becomes brighter, and the contrast of the darker tones increases. If the gamma parameter is lower than 1 then the image becomes darker, and the contrast of the brighter tones increases. If gamma is equal to 1 then no gamma correction is needed. (The 6th column contains the Gamma parameter.)
- The equation of the gamma correction is:

$$\boldsymbol{BYTE = 255 * \left(\frac{X - RANGE\_1}{RANGE\_2 - RANGE\_1}\right)^{\frac{1}{Gamma}}}$$

where,

- X is the input value, the actual calibrated value: reflectivity (R) or brightness temperature (BT),
- RANGE_1 and RANGE_2 are the limits of the range of the stretching (e.g., Red_1 and Red_2)
- Gamma is the gamma correction parameter,
- BYTE is the output value: the brightness intensity of the enhanced image (where the full range of display values is 0–255).

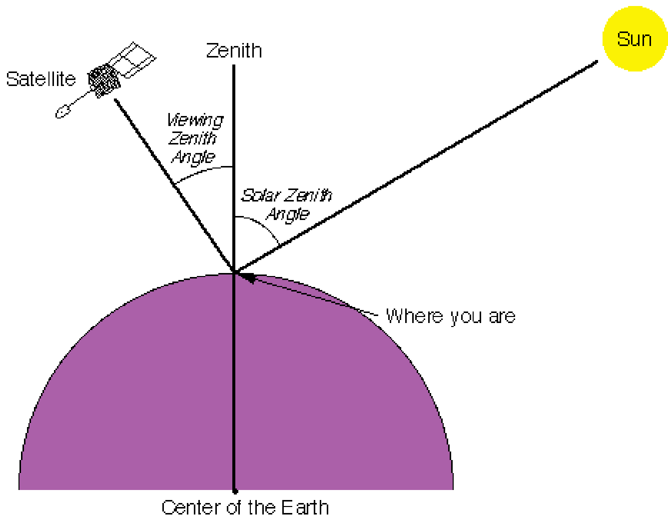
The figure below illustrates the effect of the gamma correction.



Effect of the gamma correction.

It is recommended practice to use the gamma correction parameter as in above Equation. (Ref. WMO Tokyo workshop 2017 report; https://en.wikipedia.org/wiki/Gamma_correction). It is recognised that some colleagues and software tools define the 1/Gamma exponent as gamma, i.e., they name gamma the reciprocal value of what is recommended here.

SZA explanation;



The Flexible Combined Imager (FCI) on MTG generates for some channels data simultaneously at two different spatial resolutions, e.g., IR10.5 at 1km high resolution, and at 2km normal resolution. When combining such channel data with other channels within one colour beam, to avoid artefacts, it is recommended to use imagery at identical spatial resolution, e.g., to use IR10.5 2km data in combination with IR12.3 2km data. For example, do not use 2km resolution IR12.3

(converted to 1km resolution) and 1km resolution IR10.5 data in the red component of Night Micro RGB.

However, one can use different resolution data in different colour beams. One can create an RGB from them, with no artefacts. For instance, for the Night Microphysics RGB, one can use 2 km resolution (IR12.3-IR10.5) in the red color beam (converted then to 1km resolution), and 1km data for the green (IR10.5-IR3.8) and blue (IR10.5) colour beams.

This document represents a WMO-recommended standard. Users should, however, recognize that the RGB recipes are regularly reviewed and adapted, based on operational experience, testing, and R&D. This is especially the case for the RGBs recipes based on relatively new sensors at the time of writing (2025), such as for the Flexible Combined Imager (FCI) on MTG satellites.

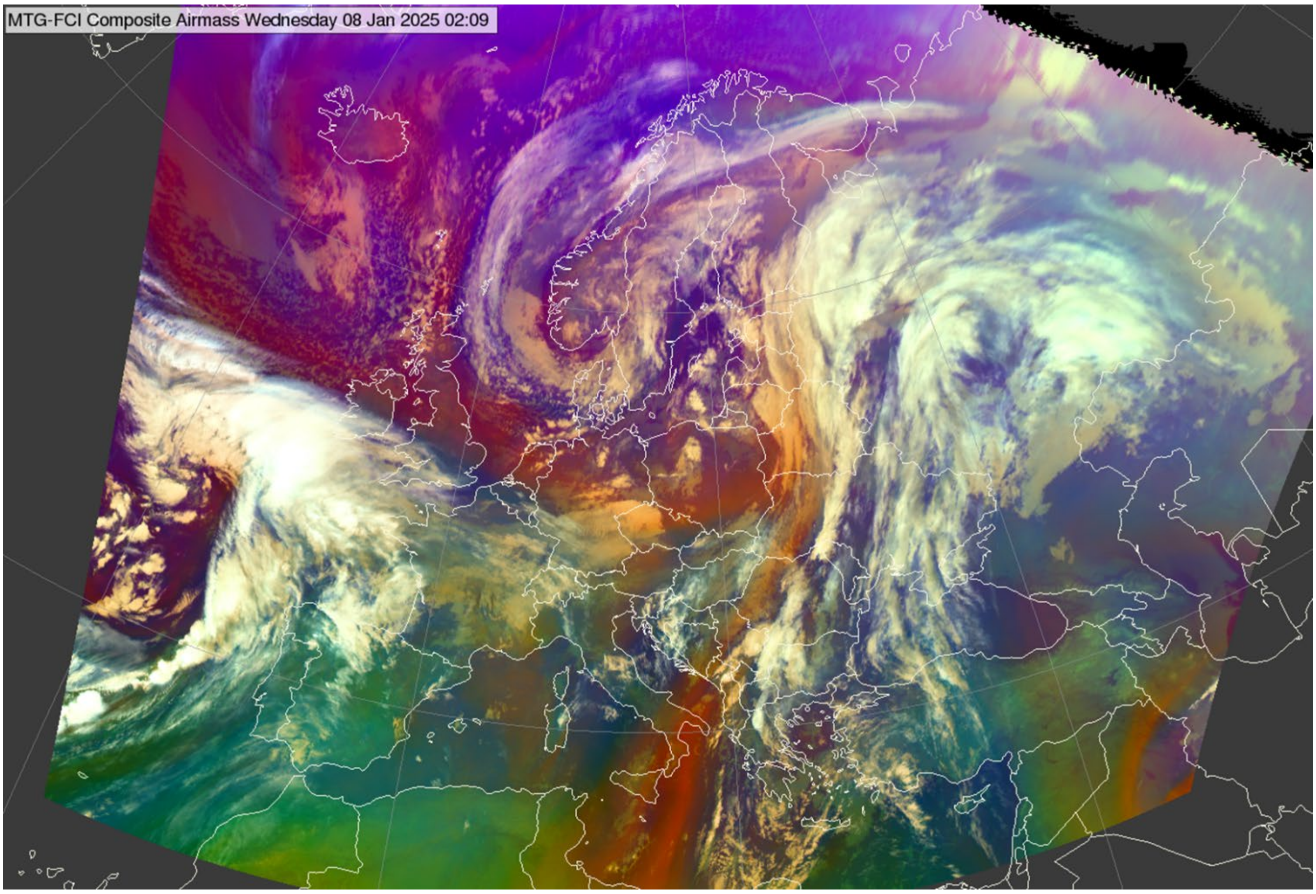
Example recipe for the Meteosat SEVIRI 24h Microphysics Cloud RGB:

Color beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-4	+2	K	1.0
Green	IR10.8 – IR8.7	0	+6	K	1.2
Blue	IR10.8	248	303	K	1.0

BASIC RGBS

Basic RGB composites are applicable globally to weather forecasting, and are recommended as core part of operational production, display systems and forecaster training. They address basic applications, such as clouds, moisture boundaries, airmasses.

Airmass RGB



FCI Airmass RGB, 8 January 2025 02:00 UTC

Main application: 24h monitoring of atmospheric dynamics
Remark: A variant for the tropics with different tuning than this Airmass RGB is listed as Special Applications RGB: Overshooting Tops RGB

MSG SEVIRI Airmass RGB

Colour beam	Channel (difference)	Range			Gamma
Red	WV6.2 – WV7.3	-25	0	K	1
Green	IR9.7 – IR10.8	-40	+5	K	1
Blue	WV6.2	243	208	K	1

MTG FCI Airmass RGB

Colour beam	Channel	Range			Gamma
Red	WV6.3 -.WV7.3	-23.8	+1.4	K	1

Green	IR9.7 - IR10.5	-39.7	+4.1	K	1
Blue	WV6.3	244.5	209.4	K	1

GOES ABI Airmass RGB

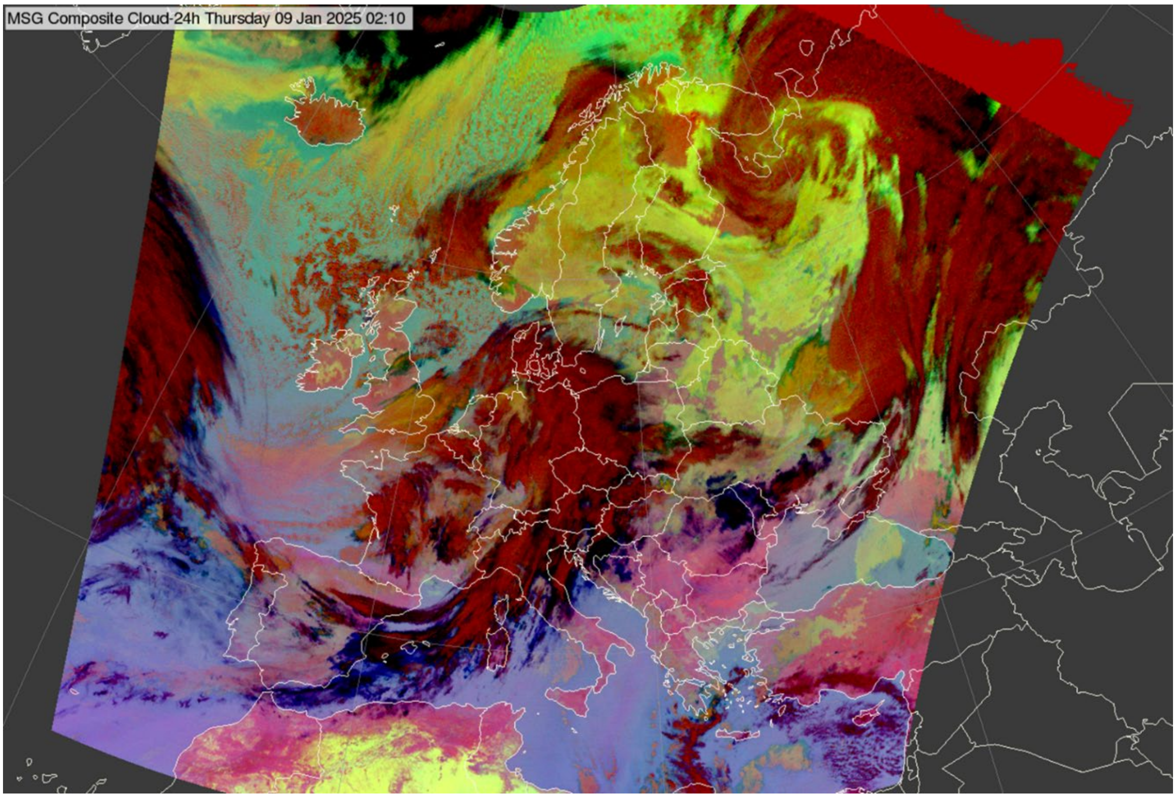
Colour beam	Channel	Range			Gamma
Red	WV6.2 - WV7.3	-26.2	+0.6	K	1
Green	IR9.6 – IR10.3	-43.2	+6.7	K	1
Blue	WV6.2	243.9	208.5	K	1

Himawari AHI Airmass RGB

Colour beam	Channel (difference)	Range			Gamma
Red	WV6.2 – WV7.3	-25.8	0	K	1.0
Green	IR9.6 – IR10.4	-41.5	+4.3	K	1.0
Blue	WV6.2	242.6	208.0	K	1.0

24-hour Microphysics Cloud RGB

Alternative name: 24-hour Microphysics RGB



MSG SEVIRI 24h Microphysics Cloud RGB, 8 January 2025 02:00 UTC

Main applications:

- 24h cloud analyses including low clouds
- 24h detection of moisture boundaries in cloud-free areas
- 24h detection of cloud phase

Remarks:

- This RGB uses the same channel (differences) as the 24h Microphysics Dust RGB, but the tuning is different. Originally tuned for Central Europe to improve the detection of low clouds (the Dust RGB does not always detect low clouds, it is tuned for dust detection.)
- The 24h Microphysics Cloud RGB performs well in twilight conditions, and in very cold conditions
- For areas where the 24h Microphysics Clouds RGB does not work well, Dust RGB could be used instead, because of its ability of moisture boundary detection (on cloud-free areas).
- Using the BTD (IR10.5-IR8.7) with FCI seems to provide less reliable cloud phase differentiation, compared to the SEVIRI data (due to a slight difference between channels IR10.8 and IR10.5).

MSG SEVIRI 24-hour Microphysics Cloud RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-4	+2	K	1.0
Green	IR10.8 – IR8.7	0	+6	K	1.2
Blue	IR10.8	248	303	K	1.0

MTG FCI 24-hour Microphysics Cloud RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.3 – IR10.5	-7.1	+2.4	K	1.0
Green	IR10.5 – IR8.7	0.2	5.2	K	1.2
Blue	IR10.5	247.8	303.1	K	1.0

Himawari AHI 24-hour Microphysics Cloud RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.4 – IR10.4	-7.5	+3	K	1.0
Green	IR11.2 - IR8.6	-0.4	+6.1	K	1.1
Blue	IR10.4	248.6	303.2	K	1.0

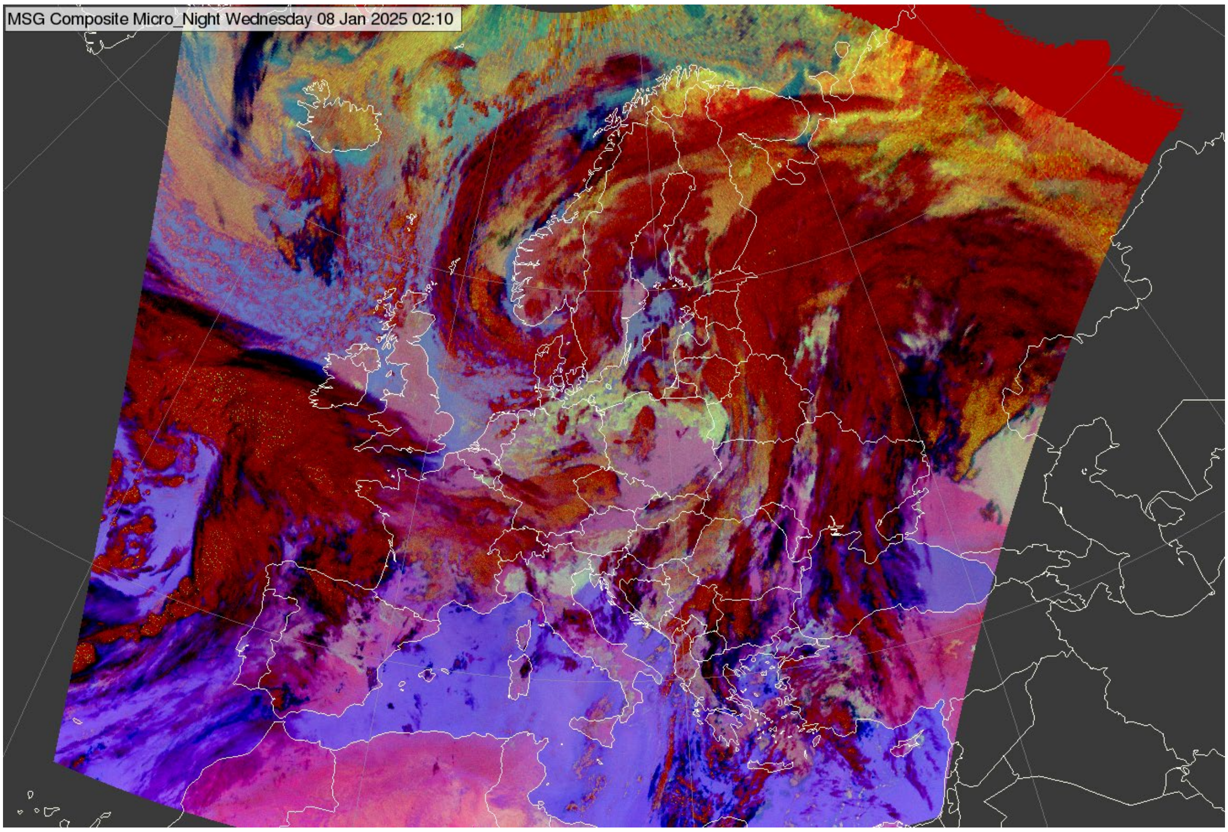
FY-4 AGRI 24-hour Microphysics Cloud RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-4	+2.0	K	1.0
Green	IR10.8 – IR8.55	0	6.0	K	1.2
Blue	IR10.8	248	303	K	1.0

GOES ABI 24-hour Microphysics Cloud RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.3 – IR10.5	-7.1	+2.4	K	1.0
Green	IR11.2 – IR8.4	0.2	5.2	K	1.2
Blue	IR10.3	247.8	303.1	K	1.0

Night Microphysics RGB



SEVIRI Night Microphysics RGB, 8 January 2025 02:00 UTC

Alternative names: Fog RGB, Fog/Low Clouds RGB
Main applications:

- nighttime detection of clouds focusing on fog/low clouds,
- nighttime detection of moisture boundaries in cloud-free areas,
- nighttime dust detection.

Remarks:

- Different tuning for tropics, see variants below.
- Performs poorly in twilight conditions, and in very cold conditions
- Present FCI recipe has a known limitation of saturation for some low clouds, to be fine-tuned in the future to mitigate for that
-

MSG SEVIRI Night Microphysics RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-4	+2	K	1
Green	IR10.8 – IR3.9	0	+10	K	1
Blue	IR10.8	243	293	K	1

MTG FCI Night Microphysics RGB

Colour beam	Channel	Range			Gamma
Red	IR12.3-10.5	-4	+2	K	1

Green	IR10.5 – IR3.8	-4	+6	K	1
Blue	IR10.5	243	293	K	1

GOES ABI Night Microphysics RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.3 - IR10.3	-6.7	+2.6	K	1.0
Green	IR10.3 – IR3.9	-3.1	+5.2	K	1.0
Blue	IR10.3	243.6	292.7	K	1.0

Himawari AHI Night Microphysics RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.4 – IR10.4	-7.5	+3	K	1.0
Green	IR10.4 – IR3.9	-2.9	+7	K	1.0
Blue	IR10.4	243.7	293.2	K	1.0

FY-4 AGRI Night Microphysics RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-4	+2	K	1.0
Green	IR10.8 – IR3.75	0	+10	K	1.0
Blue	IR10.8	243	293	K	1.0

Variants for tropical regions

Variants for tropical regions are needed to account for the larger thickness of the tropopause and the presence of generally larger amounts of humidity compared to mid-latitudes, affecting in particular the radiance signal measured in medium and long-wave IR channels. Ranges for stretching the image are therefore modified compared to the standard RGB recipe.

MSG SEVIRI Night Microphysics RGB – for tropics

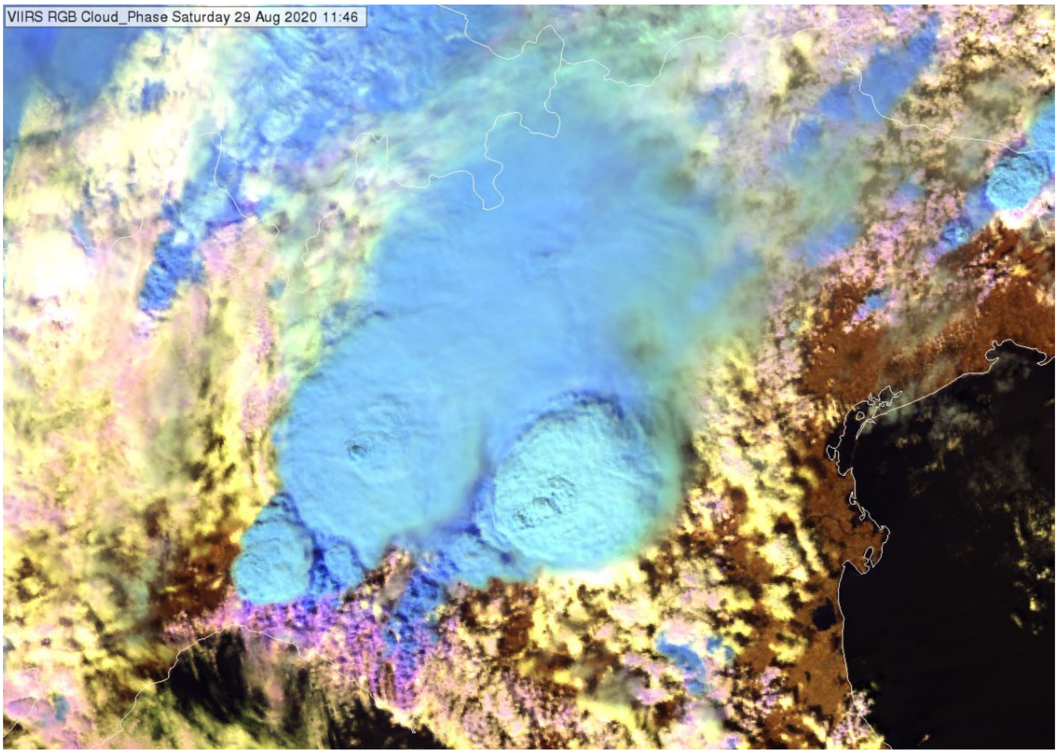
Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-4	+2	K	1
Green	IR10.8 – IR3.9	0	+5	K	1
Blue	IR10.8	273	300	K	1

MTG FCI Night Microphysics RGB – for tropics

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.5	-7.1	+2.4	K	1
Green	IR10.5 – IR3.8	-2.9	+1.1	K	1
Blue	IR10.5	273	300	K	1

Day Cloud Phase RGB

Alternative name: Cloud Phase RGB



VIIRS Cloud Phase RGB, 29 August 2020, 11:46 UTC

Main applications (daytime):

- cloud analyses, low and high cloud monitoring, cloud phase detection
- monitoring cloud top microphysics, particle size detection

Remarks:

- It provides improved cloud top microphysics information (phase and effective particle size) of thick clouds.
- The colours may be oversaturated at high solar zenith angles (near the terminator line)
- Rayleigh correction of the VIS0.64 channel is not considered necessary
- New with FCI (as of 4/2025, FCI NIR1.6 and NIR2.2 data still have biases)
- For NIR channels, the alternative range 0-70% should also be tested

MTG FCI Day Cloud Phase RGB

Colour beam	Channel (difference)	Range			Gamma
Red	NIR1.6	0	50	%	1.0
Green	NIR2.3	0	50	%	1.0
Blue	VIS0.6	0	100	%	1.0

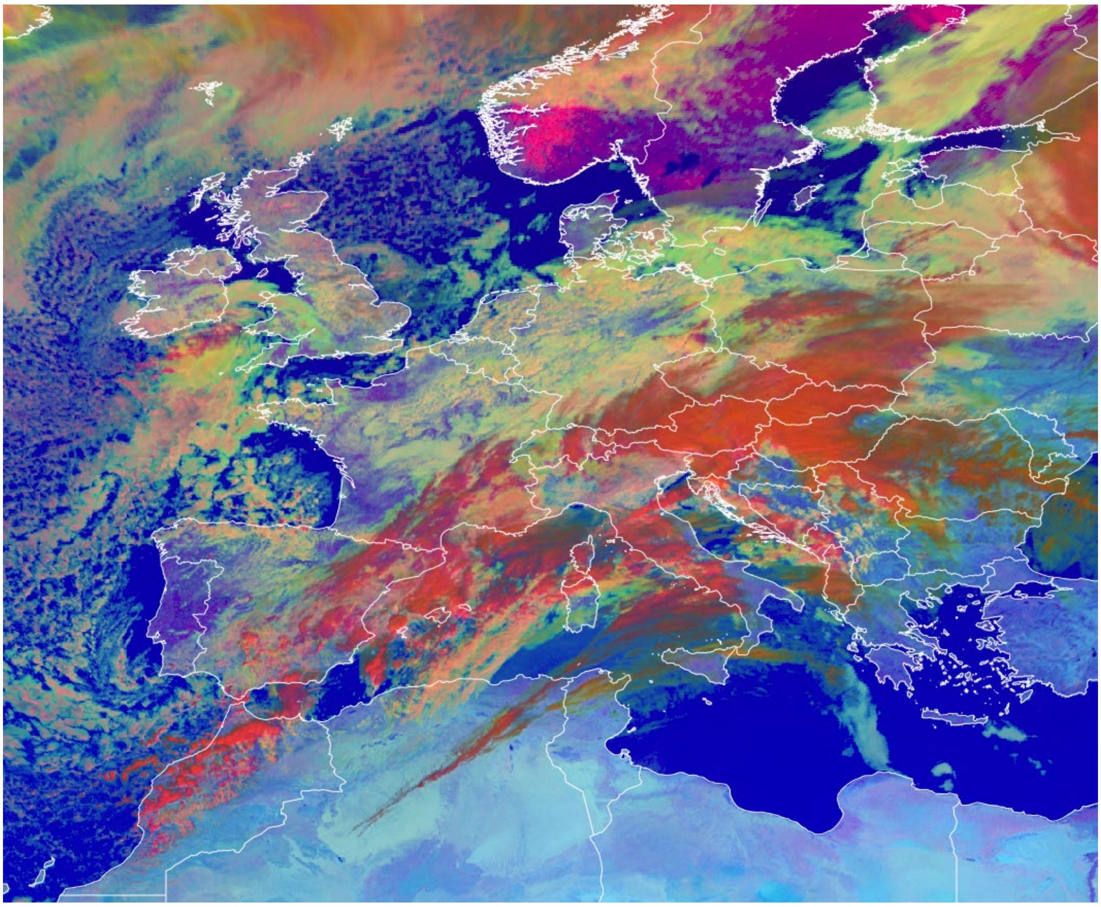
Himawari AHI Day Cloud Phase RGB

Colour beam	Channel (difference)	Range			Gamma
Red	NIR1.6	0	50	%	1.0
Green	NIR2.3	0	50	%	1.0
Blue	VIS0.64	0	100	%	1.0

GOES ABI Day Cloud Phase RGB

Colour beam	Channel (difference)	Range			Gamma
Red	NIR1.6	0	50	%	1.0
Green	NIR2.3	0	50	%	1.0
Blue	VIS0.64	0	100	%	1.0

Day Microphysics RGB



MSG SEVIRI Day Microphysics RGB, 14 March 2025, 10:45 UTC

Main applications:

- Cloud analyses
- Cloud top particle size

Remarks:

- Cloud top microphysics info is gained from the reflected part of IR3.9 (for guidance on calculation of IR3.9um reflectance component, see for instance the CIRA algorithm described here: http://nwafiles.nwas.org/digest/papers/2000/Vol24No4/Pg25-Kidder.pdf)
- For the Green beam, use of the IR3.9refl is recommended since it has highest information content on cloud microphysics. Alternatively, the 1.6µm channel can be used; see the Legacy RGB section for Day Microphysics RGB (with NIR1.6). Alternative use of NIR2.25 channel should be explored.
- It contains temperature information as well (blue component).
- The Day Cloud Phase RGB and the Day Microphysics RGBs complement each other.

MSG SEVIRI Day Microphysics RGB

Color beam	Channel	Range			Gamma
Red	VIS0.8	0	100	%	1.0
Green	IR3.9refl	0	60	%	2.5
Blue	IR10.8	203	323	K	1.0

MTG FCI Day Microphysics RGB

Color beam	Channel	Range			Gamma
Red	VIS0.8	0	100	%	1.0
Green	IR3.8refl	0	60	%	2.5
Blue	IR10.5	203	323	K	1.0

Himawari AHI Day Microphysics RGB

Color beam	Channel	Range			Gamma
Red	NIR0.86	0	102	%	0.95
Green	IR3.9refl	2	82	%	2.6
Blue	IR10.4	203.5	303.2	K	1.0

GOES ABI Day Microphysics RGB

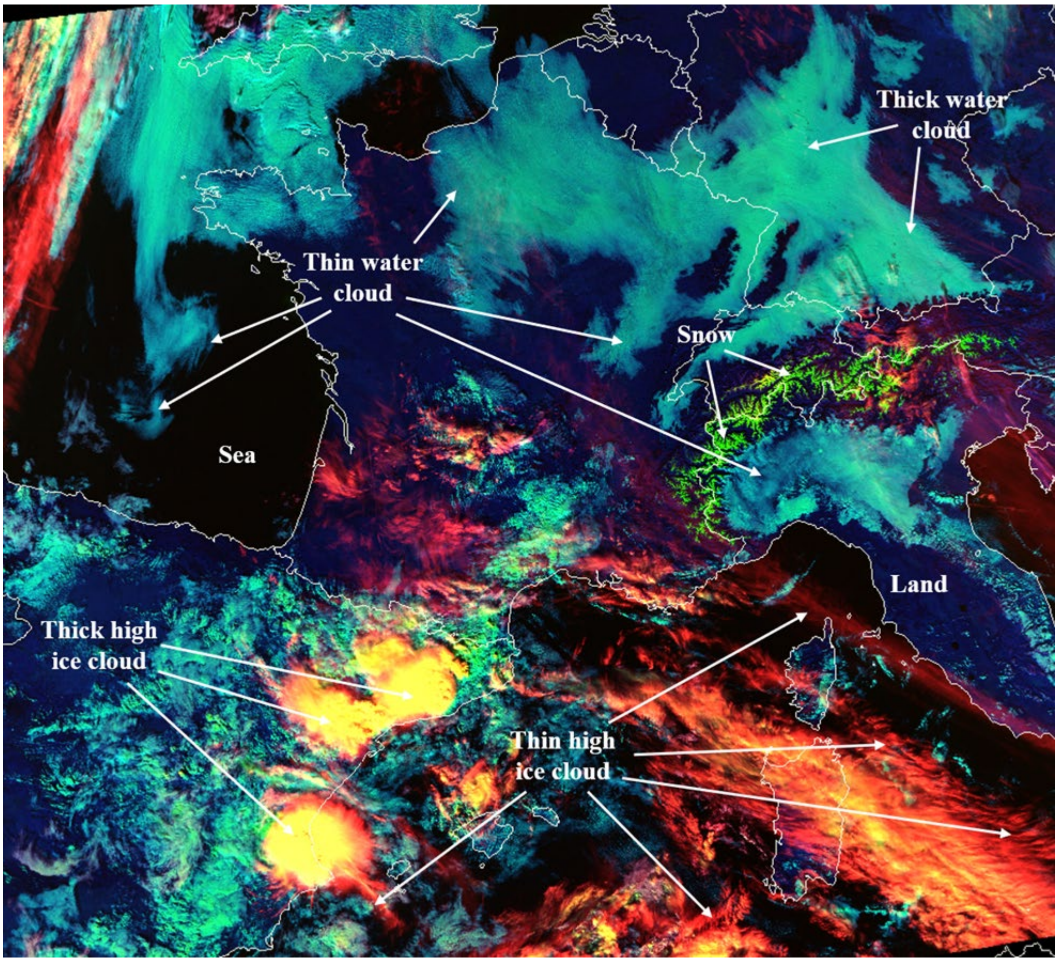
Color beam	Channel	Range			Gamma
Red	VIS0.8	0	100	%	1.0
Green	IR3.9refl	0	60	%	2.5
Blue	IR10.3	203	323	K	1.0

FY-4 AGRI Day Microphysics RGB

Color beam	Channel	Range			Gamma
Red	VIS0.825	0	100	%	1.0
Green	IR3.75refl	0	60	%	2.5
Blue	IR10.8	203	323	K	1.0

Day Cloud Type RGB

Alternative name: Cloud Type RGB



VIIRS Cloud Type RGB with interpretation, 15 November 2018, 12:31 UTC.

Main applications (daytime):

- detection of high-level thin cirrus clouds
- detection of high-level thin aerosol plumes
- identification of cloud types
- can provide some information about areas with dry airmasses.

Remarks:

- contains information on cloud height
- Similar to Day Cloud Type Distinction RGB (which contains IR10.x temperature information on Red)
- new with FCI

MTG FCI Day Cloud Type RGB

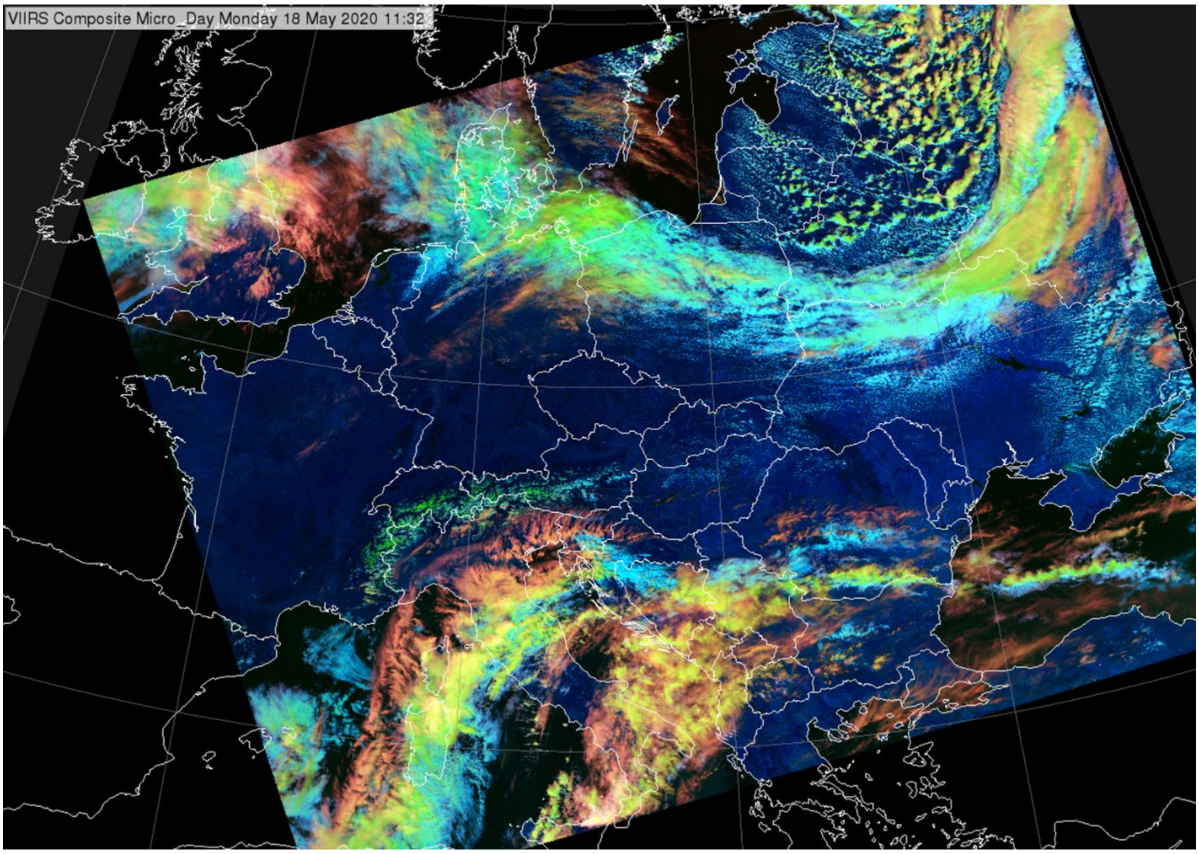
Colour beam	Channel	Range			Gamma
Red	NIR1.37	0	10	%	1.5
Green	VIS0.64	0	80	%	0.75
Blue	NIR1.6	0	80	%	1

GOES ABI Day Cloud Type RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.37	0	10	%	1.5
Green	VIS0.64	0	78	%	1
Blue	NIR1.6	0	59	%	1

Variant: Day Cloud Type Distinction RGB

Alternative name: Day Cloud Phase Distinction for AHI and ABI



VIIRS Day Cloud Type Distinction RGB, 18 May 2020, 11:28 UTC

Main application (daytime):

- General cloud analyses
- Detection of convective initiation

Remarks:

- It uses almost the same channels as Day Microphysics RGB, but with different colour assignment. It therefore looks very differently but the information content is almost the same (except that Day Microphysics has particle size information through IR3.8refl on green beam).
    - In case of Day Cloud Type Distinction RGB the focus is on cloud optical thickness (and temperature),
    - In case of Day Microphysics RGB the focus is on cloud top microphysics (and optical thickness).
- The alternative name “Day Cloud Phase Distinction” is confusing (but in broad use by some forecasters) since this RGB is very different from the “Day Cloud Phase RGB” but close to the “Day Cloud Type RGB” therefore it is recommended to here use the term “Day Cloud Type Distinction RGB”
- Its typical colours are close to the colours of the Cloud Type RGB, while the interpretation is partly similar, partly different. This may be confusing if both are used. The Cloud Type RGB uses the NIR1.3 in the Red, whereas this RGB uses the IR10.X from ABI and AHI.

GOES ABI Day Cloud Phase Distinction RGB

Colour beam	Channel	Range			Gamma
Red	IR10.3	280.7	219.6	K	1.0
Green	VIS0.64	0	78	%	1.0
Blue	NIR1.6	1	59	%	1.0

Himawari AHI Cloud Phase Distinction RGB

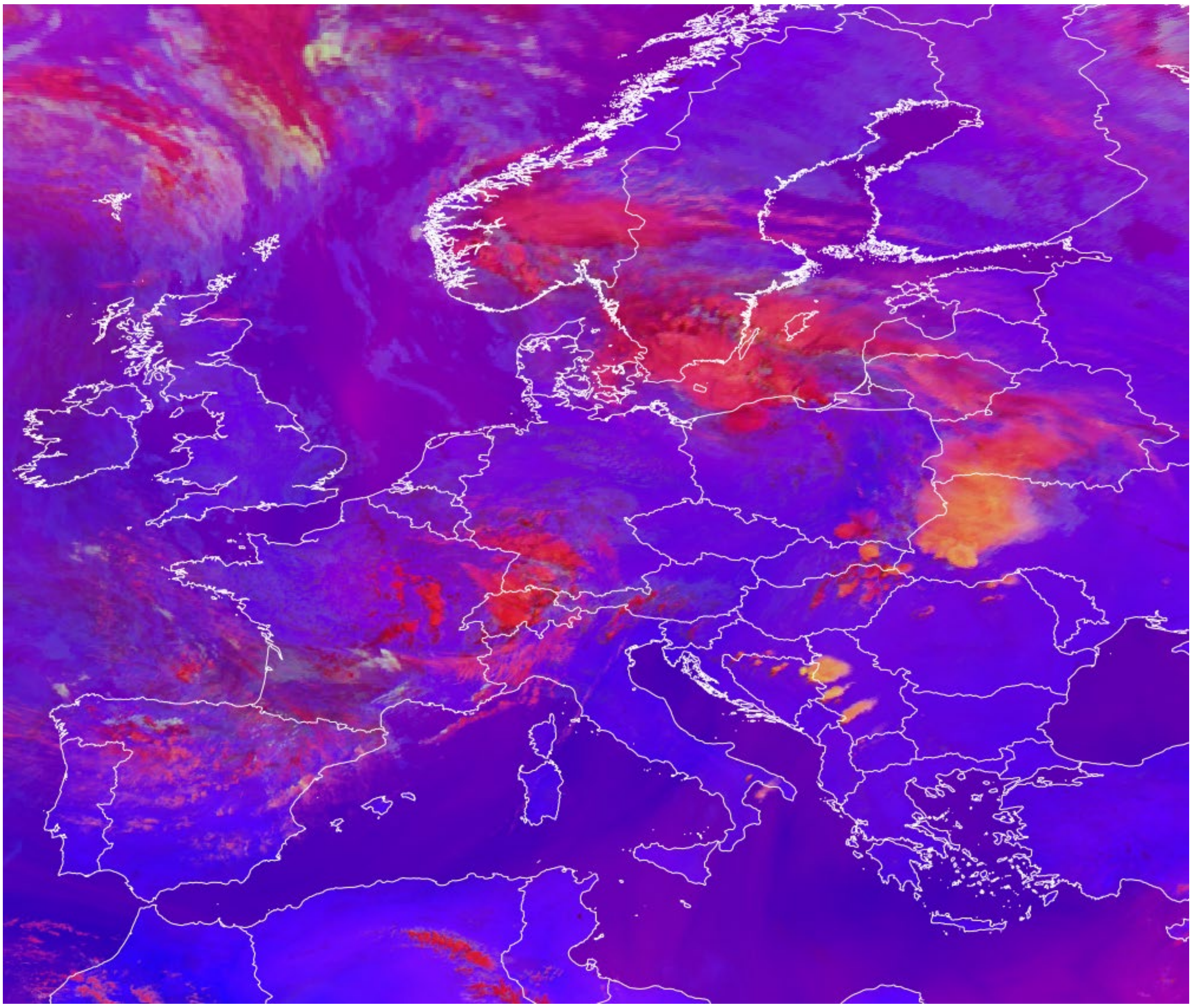
Colour beam	Channel	Range			Gamma
Red	IR10.4	280.7	219.6	K	1.0
Green	VIS0.64	0	85	%	1.0
Blue	NIR1.6	1	50	%	1.0

SPECIAL APPLICATIONS RGBS

Special Applications RGBs are relevant to specific contexts and applications (detection of fires, dust, ash, SO2, snow), recommended for operational production, display and training on a context-dependent, on-demand basis.

Day Severe Storms RGB

Alternative names: Severe Storms RGB, Convection RGB, Severe Convection RGB



SEVIRI Severe Storms RGB, 22 June 2024, 13:30 UTC

Main applications (daytime):

- detection of small ice particles (on top of the high clouds)
- monitoring convection, high-level lee clouds, very polluted high clouds, clouds with high cloud base, and supercooled water clouds

Remarks:

- Focuses on high ice clouds
- The green component (mostly responsible for the levels of yellow shades) provides information on cloud top particle size, but the signal slightly depends on cloud top temperature as well.
- The colour contrast between ice clouds covered by large/small (and/or very cold) particles is excellent.
- Its interpretation of ice particle size distribution is rather straightforward. The Cloud Phase RGB can partly replace this RGB for ice cloud particle size distinction (though not so prominent) without cloud top temperature dependence. However, in some cases they complement each other, e.g. for detecting supercooled cloud droplets.
- Different tuning for tropical regions, see variants below.
- Due to noise appearing at very low temperatures in the FCI IR3.8 channel, green noisy pixels may show up across coldest cloud tops

MSG SEVIRI Day Severe Storms RGB

Colour beam	Channel difference	Range			Gamma
Red	WV6.2 – WV7.3	-35	+5	K	1
Green	IR3.9 – IR10.8	-5	+60	K	0.5
Blue	NIR1.6 – VIS0.6	-75	+25	%	1

MTG FCI Day Severe Storms RGB

Colour beam	Channel	Range			Gamma
Red	WV6.3 - WV7.3	-30	0	K	1
Green	IR3.8 – IR10.5	0	55	K	0.5
Blue	NIR1.6 - VIS0.6	-70	+20	%	1

GOES ABI Day Severe Storms RGB

Colour beam	Channel	Range			Gamma
Red	WV6.2 - WV7.3	-35	+5	K	1
Green	IR3.9 – IR10.3	-5	+60	K	0.5
Blue	NIR1.6 - VIS0.64	-75	+25	%	1

Himawari AHI Day Severe Storms RGB

Colour beam	Channel (difference)	Range			Gamma
Red	WV6.2 - WV7.3	-36.0	+5	K	1.0
Green	IR3.9 - IR10.4	-1.0	+61.0	K	0.5
Blue	NIR1.6 - VIS0.64	-80	+26	%	0.95

FY-4 AGRI Day Severe Storms RGB

Colour beam	Channel difference	Range			Gamma
Red	WV6.2 – WV6.95	-35	+5	K	1
Green	IR3.75 – IR10.8	-5	+60	K	0.5
Blue	NIR1.6 – VIS0.65	-75	+25	%	1

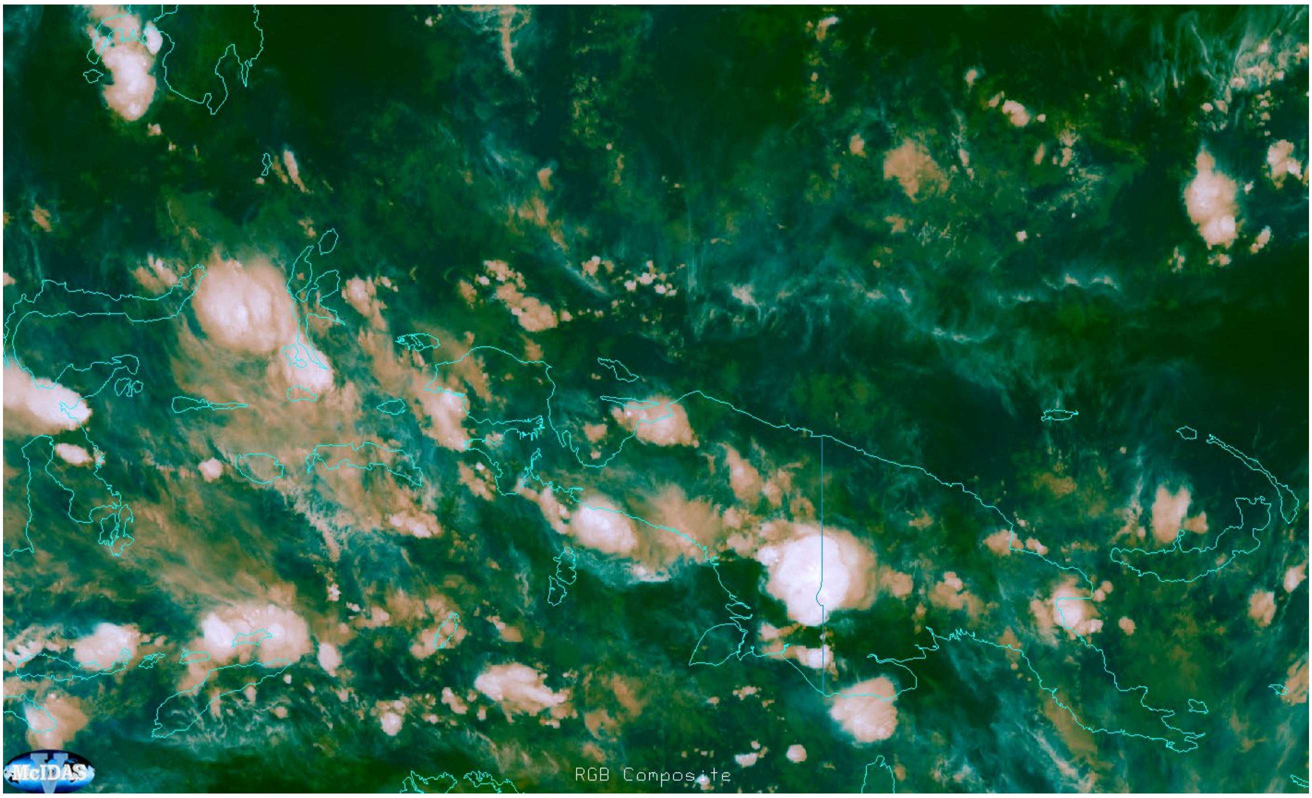
Variants for tropical regions

Variants for tropical regions are needed to account for the larger thickness of the tropopause and the presence of generally large amounts of humidity compared to mid-latitudes, affecting in particular the radiance signal measured in medium and long-wave IR channels. Ranges for stretching the image are therefore modified compared to the standard RGB recipe.

MSG SEVIRI Day Severe Storms RGB – for tropics

Colour beam	Channel difference	Range			Gamma
Red	WV6.2 – WV7.3	-35	+5	K	1
Green	IR3.9 – IR10.8	-5	+75	K	0.33
Blue	NIR1.6 – VIS0.6	-75	+25	%	1

Overshooting Tops RGB



Himawari-9 AHI, Overshooting Tops RGB, 17 March 2025, 13:30 UTC

Main application: 24h monitoring of overshooting tops

Remarks:

- This RGB is a variant of the basic Airmass RGB where the ranges, and the channel difference in the Red colour beam is different

MSG SEVIRI Overshooting Tops RGB

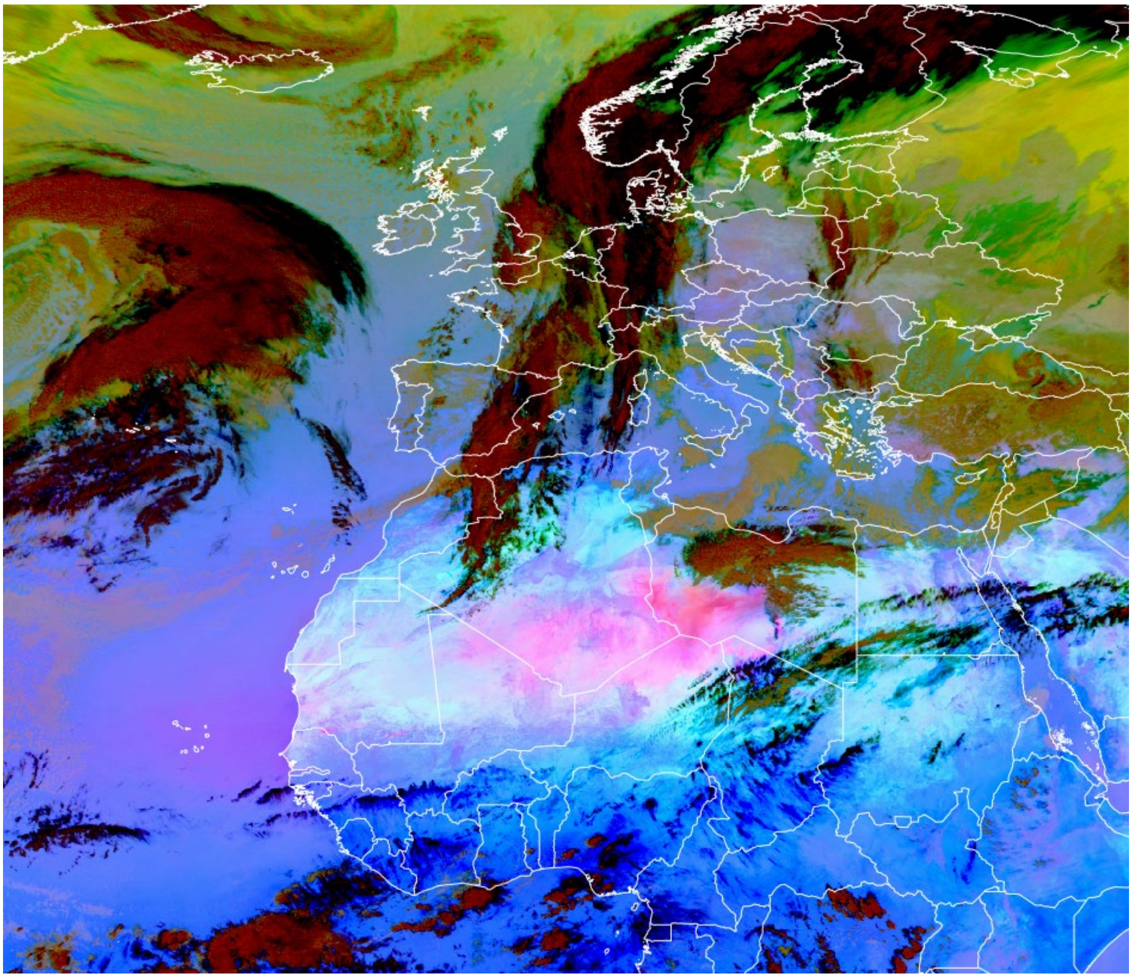
Colour beam	Channel (difference)	Range			Gamma
Red	WV6.2 – IR10.8	-25	+5	K	1
Green	IR9.7 – IR10.8	-30	+25	K	1
Blue	WV6.2	243	190	K	1

MTG FCI Overshooting Tops RGB

Colour beam	Channel (difference)	Range			Gamma
Red	WV6.3 – IR10.5	-23.8	+6.4	K	1
Green	IR9.7 – IR10.5	-29.9	+23.6	K	1
Blue	WV6.3	244.5	191.4	K	1

24-hour Microphysics Dust RGB

Alternative name: Dust RGB



SEVIRI Dust RGB, 22 February 2025, 12:45 UTC

Main applications:

- 24h dust detection
- 24h detection of moisture boundaries on cloud-free areas
- 24h detection of clouds (low clouds might be missed)
- 24h detection of cloud phase

Remarks:

- Same channel combinations on the RGB components as for 24-hour Microphysics Cloud RGB, only different ranges are applied
- Low clouds are not always detected; this may cause moisture boundary false detection.
- In cases where the 24h Microphysics Cloud RGB saturates or does not provide sufficient contrast, the Dust RGB should be used because it is based on the same channels but uses a different tuning.

MSG SEVIRI Dust RGB

Color beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-4	+2	K	1.0
Green	IR10.8 – IR8.7	0	+15	K	2.5
Blue	IR10.8	261	289	K	1.0

MTG FCI Dust RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 – IR10.8	-7.1	+2.4	K	1.0
Green	IR10.5 – IR8.7	0.2	12.7	K	2.5
Blue	IR10.8	260.9	289.0	K	1.0

GOES ABI Dust RGB

Color beam	Channel (difference)	Range			Gamma
Red	IR12.3 – IR10.3	-6.7	+2.6	K	1.0
Green	IR11.2 – IR8.4	-0.5	+20	K	2.5
Blue	IR10.3	261.2	288.7	K	1.0

Himawari AHI Dust RGB

Color beam	Channel (difference)	Range			Gamma
Red	IR12.4 –IR10.4	-7.5	+3	K	1.0
Green	IR11.2 – IR8.6	-0.5	+15.0	K	2.2
Blue	IR10.4	261.5	289.2	K	1.0

FY-4 AGRI Dust RGB

Color beam	Channel (difference)	Range			Gamma
Red	IR12.0 –IR10.8	-4.0	+2	K	1.0
Green	IR10.8 – IR8.55	0.0	+15.0	K	2.5
5Blue	IR10.8	261.0	289.0	K	1.0

Variant for users with Colour Vision Deficiency (CVD)

Full name: 24h Microphysics CVD Dust RGB

Alternative name: CVD Dust RGB

Main applications:

- 24h dust detection

- 24h detection of moisture boundaries on cloud-free areas
- 24h detection of clouds (low clouds might be missed)
- 24h detection of cloud phase

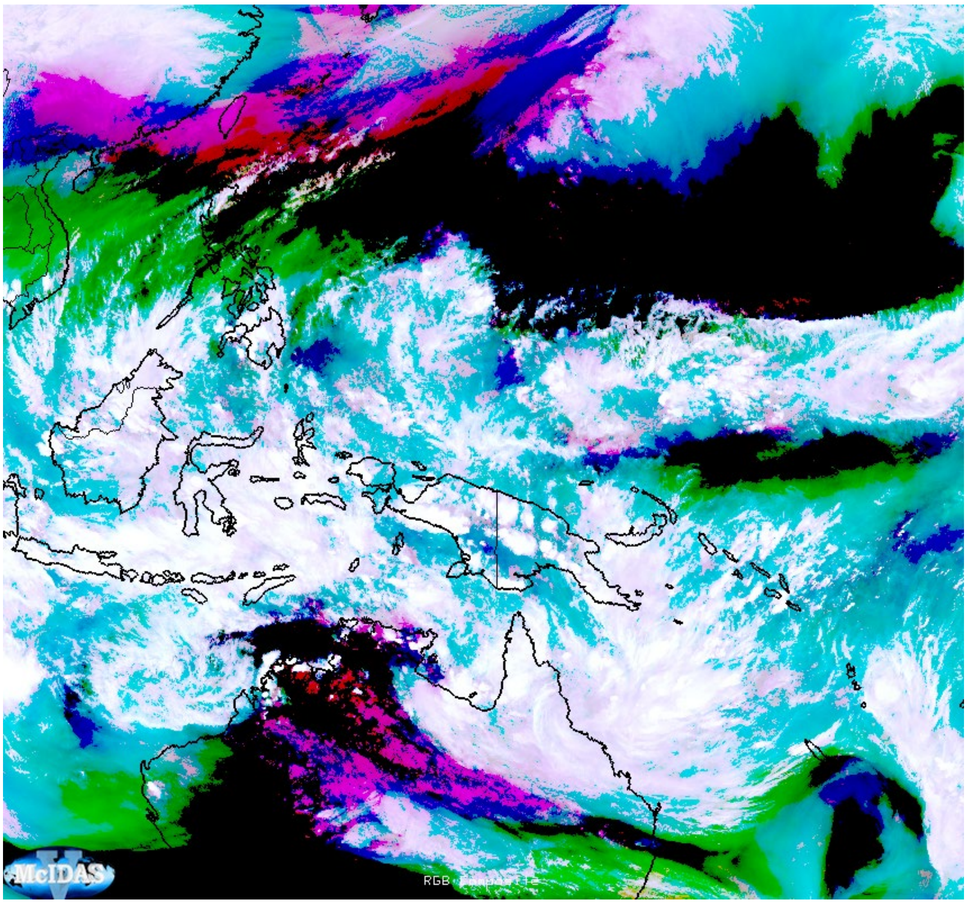
Remarks:

- For users with colour vision deficiency (CVD)
- Low clouds are not always detected, this may cause moisture boundary false detection.
- In cases where the 24h Microphysics Cloud RGB saturates or does not provide sufficient contrast, the Dust RGB should be used because it is based on the same channels but uses a different tuning.

GOES ABI CVD Dust RGB

Color beam	Channel (difference)	Range			Gamma
Red	IR10.3 – IR8.4	+6	-0.5	K	0.67
Green	IR12.3-IR10.3	-6	+2.5	K	0.67
Blue	IR10.3	233	313	K	1.0

Simple Water Vapour RGB



AHI9, Simple WV RGB, 17 March 2025, 09:00 UTC

Main applications:

- Analysis of atmospheric water vapour distribution for individual levels excluding cloud areas

Remarks:

- Considered useful for users with Colour Vision Deficiency (CVD)
- Colour components providing features on different atmospheric levels such as the jet stream, upper cold lows, moisture return, conveyor belts and gravity waves (mountain waves)
- Applicability day and night thanks to infrared image composition
- Display of convective initiation as pinkish-whitish cloud
- Lack of clarity in features such as low-level cloud/fog
- Saturation of shading for high-level clouds (whitish)
- Color shading effects from satellite viewing angle, particularly in limb areas (limb cooling effect)

GOES ABI Simple Water Vapour RGB

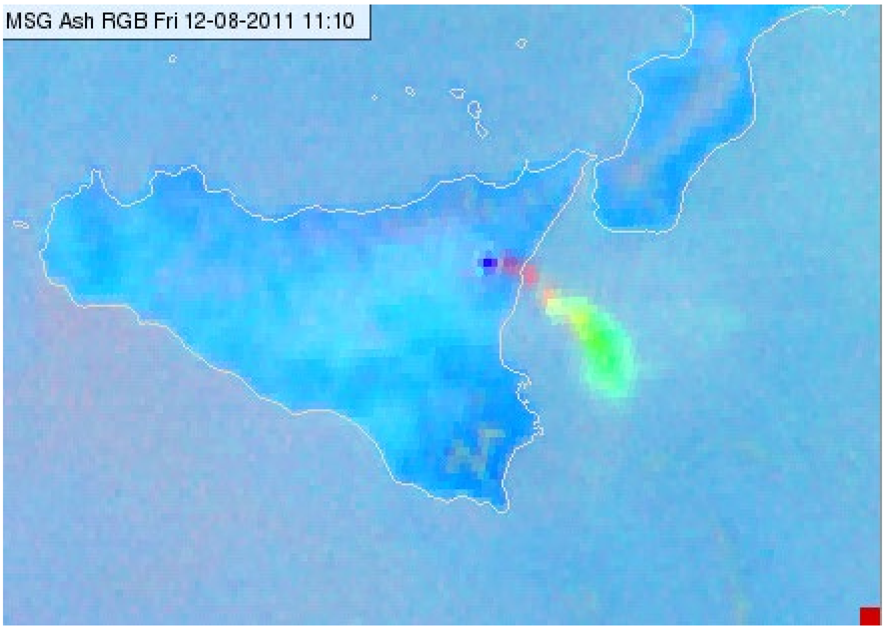
Colour beam	Channel	Range			Gamma
Red	IR10.3	279.0	202.3	K	10.0
Green	WV6.2	242.7	214.7	K	5.5
Blue	WV7.3	261.0	245.1	K	5.5

Himawari AHI Simple Water Vapour RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR10.4	279.0	202.3	K	10.0
Green	WV6.2	242.7	214.7	K	5.5
Blue	WV7.3	261.0	245.1	K	5.5

24-hour Microphysics Ash RGB

Alternative name: Ash RGB



SEVIRI Ash RGB, 12 August 2011, 11:10 UTC

Main applications (24h):

- Detection volcanic ash
- Detection volcanic SO_2 gas plume
- Not working on limb

- Remark: Same channel combinations on the RGB components as for 24-hour Microphysics Cloud RGB, only different ranges are applied

MSG SEVIRI Ash RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 - IR10.8	-4	+2	K	1
Green	IR10.8 – IR8.7	-4	+5	K	1
Blue	IR10.8	243	303	K	1

MTG FCI Ash RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.0 - IR10.8	-7.1	+2.4	K	1
Green	IR10.8 – IR8.7	-3.2	+4.4	K	1
Blue	IR10.8	242.8	303.1	K	1

GOES ABI Ash RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR12.3 – IR10.3	-6.7	+2.6	K	1.0

Green	IR11.2 – IR8.4	-6.0	+6.3	K	1.0
Blue	IR10.3	243.6	302.4	K	1.0

Himawari AHI Ash RGB

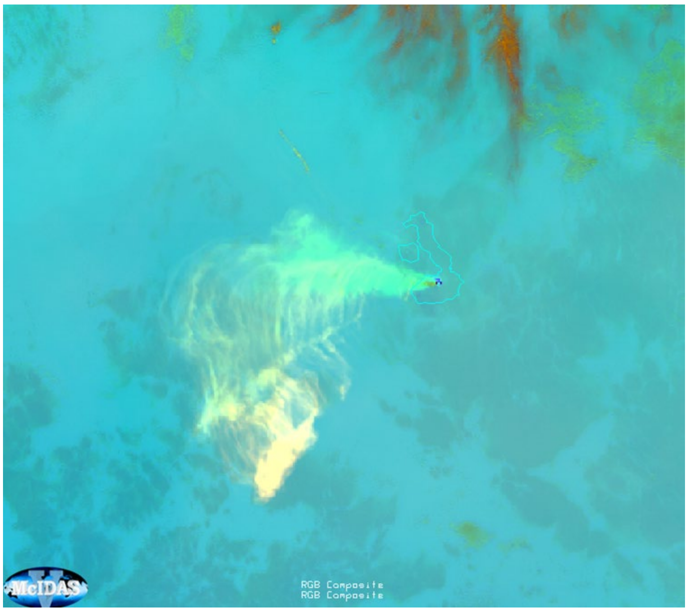
Colour beam	Channel (difference)	Range			Gamma
Red	IR12.4 - IR10.4	-7.5	+3	K	1.0
Green	IR11.2 - IR8.6	-1.6	+4.9	K	1.2
Blue	IR10.4	243.6	303.2	K	1.0

FY-4 AGRI Ash RGB

Colour beam	Channel(difference)	Range （K）			Gamma
R	12-10.8	-4	+6	K	1
G	10.8-8.55	-4	+5	K	1
B	10.8	243	303	K	1

Volcanic Emissions (SO2 and Ash) RGB

Alternative name: SO2 and Ash RGB
Previous name: SO2 RGB



GOES-16, SO2 RGB, 27 June 2018, 01:30 UTC, Sierra Negra volcano, Ecuador

Main applications (24h):

- Detection of volcanic SO_2 gas plume and volcanic ash
- Improved detection compared to Ash RGB. Both the red and the green components provide information on SO2 gas plume.
- It may provide some information on the SO2 gas plume height as well.
- Also, detection of dust and ash.
- Note that FCI recipe for this RGB is direct copy of ABI recipe (further validation needed)

MTG FCI SO2 and Ash RGB

Colour beam	Channel (difference)	Range			Gamma
Red	WV6.3 – WV7.3	-4.0	+2.0	K	1.0
Green	IR10.3 – IR8.4	-4.0	+5.0	K	1.0
Blue	IR10.3	243	303	K	1.0

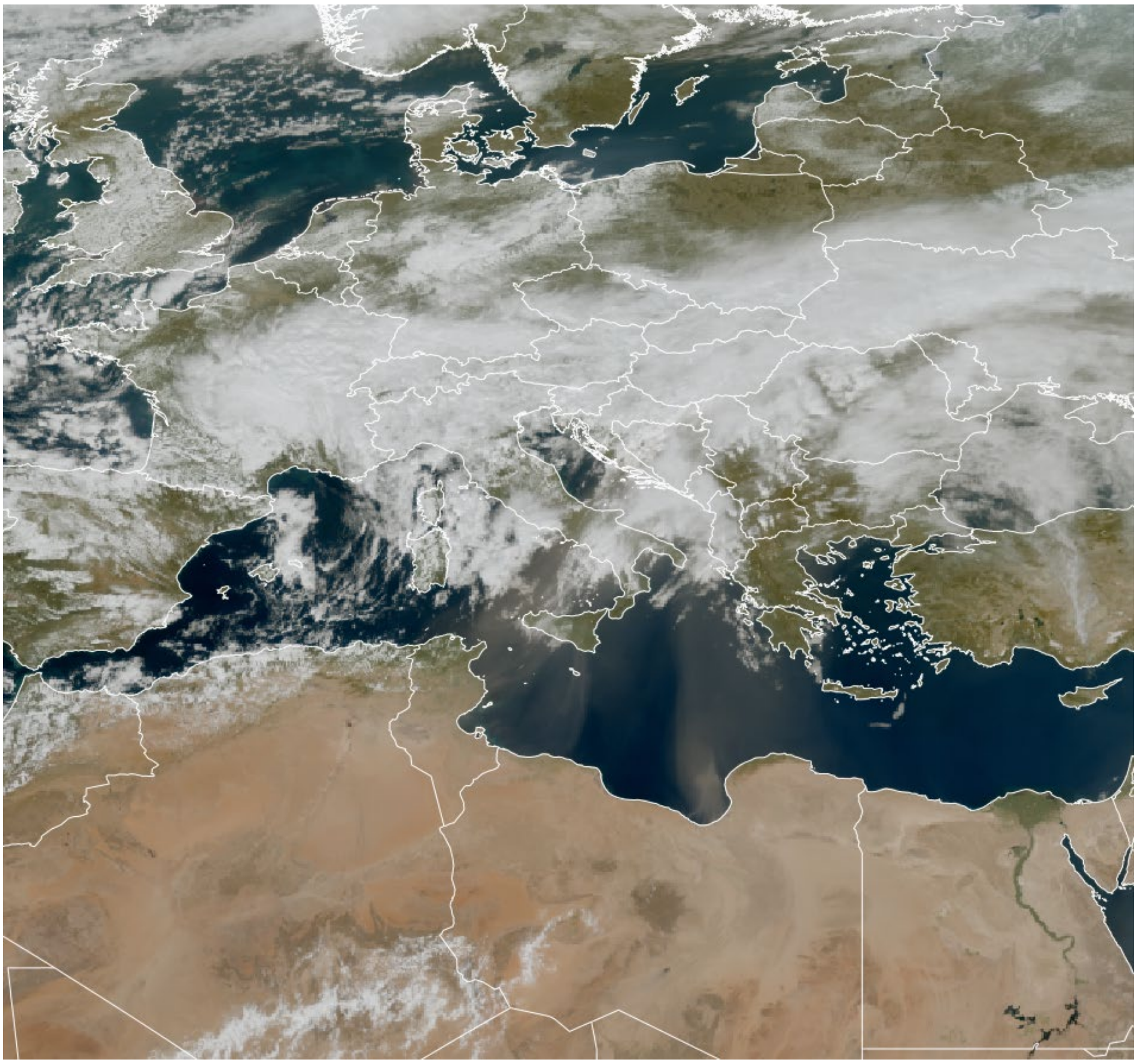
GOES ABI SO2 and Ash RGB

Colour beam	Channel (difference)	Range			Gamma
Red	WV6.9 – WV7.3	-4.0	+2.0	K	1.0
Green	IR10.3 – IR8.4	-4.0	+5.0	K	1.0
Blue	IR10.3	243	303	K	1.0

Himawari AHI SO2 and Ash RGB

Colour beam	Channel (difference)	Range			Gamma
Red	WV6.9 – WV7.3	-5.0	6.0	K	1.0
Green	IR11.2 – IR8.6	-5.9	5.1	K	0.85
Blue	IR10.4	243.6	303.2	K	1.0

True Colour RGB



FCI True Colour RGB, 15 March 2025, 12:20 UTC

Main applications (daytime):

- Detection of aerosols (especially over water bodies), with slight indication on the aerosol type (not as distinct as e.g. with Day Land Cloud RGB)
- Monitoring of water bodies (turbidity, algae blooms, chlorophyll, white caps, etc.)
- Detection of clouds and snow (including frost or rime) - all in white/grey shades with hints on cloud thickness, but no classification of cloud phase or particle size
- Includes (usually hybrid) vegetation signal (e.g. detection of burnt areas)
- Communication to general public (media items), for its natural colours closest to human eye perception

Remarks:

- True Colour RGB is sometimes used without Rayleigh correction (or even sun zenith angle (SZA) correction) itofor example when there is limited processing capability, or when wanting to preserve the realistic Earth space view..
- Reflectance values are typically scaled log-linearly to replicate the response of the human eye. F
- Detection of aerosols works very well with this RGB, especially under certain conditions (strong forward/backward scattering over water bodies)

General recipe for all sensors is:

True Colour RGB

Colour beam	Channel	Range			Gamma
Red	VIS0.6	0	100	%	NA
Green	VIS0.5	0	100	%	NA
Blue	VIS0.4	0	100	%	NA

Himawari AHI True Colour RGB

Steps for creating the AHI True Colour RGB (source: Miller et al., 2016):

- The VIS0.47, VIS0.51, VIS0.64 and NIR0.86 reflectance values are normalised by the atmospheric transmittance along the Sun–Earth–satellite photon path (the suggested solar zenith angle correction method follows Li and Shibata (2006)).
- Rayleigh correction of VIS0.47, VIS0.51, VIS0.64 and NIR0.86 channel reflectance is typically applied to get a clearer imagery with larger contrasts. To avoid overcorrection at high satellite and/or solar zenith angles (limb and terminator), where high/thick clouds significantly reduce the photon optical path through the atmosphere compared to clear sky, the Rayleigh component is reduced by a factor scaled by the cloud-top height, using e.g. the 10.35-μm brightness temperature or 0.6-μm reflectance as a proxy for cloud height. For more natural-looking imagery, that better resembles the Earth as seen from space, Rayleigh correction should not be applied.
- Since the AHI green channel does not capture the spectral reflectance peak of chlorophyll (vegetation) around 0.55 microns, VIS0.51and NIR0.8 are combined/blended to boost the green vegetation signal. This is called the 'hybrid green' band (G_H): $G_H = (1\text{-}F) * R\_510 + F * R\_856$, $F \sim 0.07$ (experimental)
- Finally, the corrected reflectance values are scaled log-linearly to enhance the imagery and replicate the response of the human eye (rather than typical gamma or piecewise linear stretching).

GOES ABI True Colour RGB

ABI does not measure in the green visible region. For generating a True Colour RGB, the green visible channel data is simulated using other ABI channel data, and Himawari AHI data. The simulation method is based on AHI data and is described in Miller et al. (2012). Otherwise, the algorithm is similar to the algorithm described above (Miller et al., 2016).

MTG FCI True Colour RGB

Similar to the AHI True Colour RGB method: Rayleigh correction of VIS0.4, VIS0.5 andVIS0.6 channel reflectance values. Since FCI also miss the reflectance peak of chlorophyll, a hybrid green signal is derived using the VIS0.5 and NIR0.8 channel reflectance values to enhance the vegetation signal. However, unlike AHI's use of a fixed NIR0.8 fraction (F ~ 0.07), this approach calculates pixel-level values dynamically based on the Normalized Difference Vegetation Index (NDVI).

Fire Temperature RGB

ABI Fire Temperature RGB, 9 September 2020, 21:00 UTC

Main applications (24h):
- Fire detection
- Qualitative information on fire intensity

Remarks:
- Some recipes have two variants:
  - A recipe with radiance values (preferred if available) and
  - A recipe with not normalized reflectance and brightness temperature values.
- To see small, moderately hot fires, usage of high spatial resolution channel data is recommended (e.g., FCI NIR2.2 at 0.5km resolution).
- At night, information on clouds, smoke and the background land surface is generally unavailable.
- During daytime, burnt areas, water and ice clouds are seen. Smoke is usually not seen except extremely dense smoke and mainly over the sea (see example image below). Many other features are not well depicted.
- Very warm surfaces may mask the fire signals (too strong red signal)
- Sun glint can be an issue.

SNPP/NOAA-20/21 VIIRS Fire Temperature RGB

Colour beam	Channel	Range			Gamma
Red	IR3.7	0	3.5	W m^{-2} sr^{-1} μm^{-1}	1

Green	NIR2.25	0	35.0	W m^{-2} sr^{-1} μm^{-1}	1
Blue	NIR1.6	0	85.0	W m^{-2} sr^{-1} μm^{-1}	1

MTG FCI Fire Temperature RGB

Colour beam	Channel	Range			Gamma
Red	IR3.8	273 0	333* 5.1	K W m^{-2} sr^{-1} μm^{-1}	0.4 1
Green	NIR2.25	0 0	100 ** 17.7	% W m^{-2} sr^{-1} μm^{-1}	1 1
Blue	NIR1.6	0 0	75 ** 22.0	% W m^{-2} sr^{-1} μm^{-1}	1 1

* it is recommended that users in warmer, arid areas adjust the high end of the range to 343 K to avoid saturation in non-fire conditions.

* *The NIR2.2 and NIR1.6 calibrated values should NOT be normalized with sun zenith angle, i.e., no sun zenith angle correction be applied for these channels. In case of fires, thermal radiation is measured in NIR2.2 and NIR1.6 channels.

GOES ABI Fire Temperature RGB

Colour beam	Channel	Range			Gamma
Red	IR3.9	273	333*	K	0.4
Green	NIR2.2	0	100 **	%	1
Blue	NIR1.6	0	75 **	%	1

* it is recommended that users in warmer, arid areas adjust the high end of the range to 343 K to avoid saturation in non-fire conditions.

* *The NIR2.2 and NIR1.6 calibrated values should NOT be normalized with sun zenith angle, i.e., no sun zenith angle correction be applied for these channels. In case of fires, thermal radiation is measured in NIR2.2 and NIR1.6 channels.

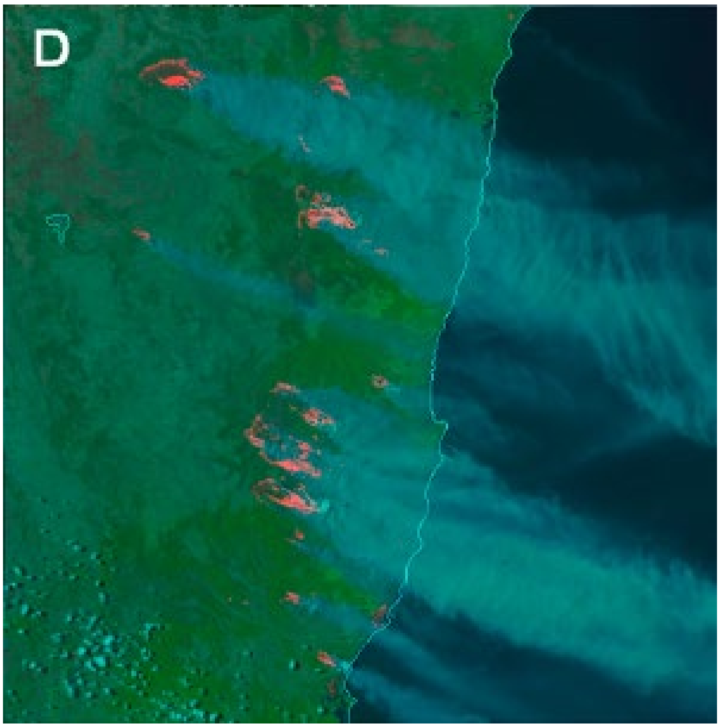
Himawari AHI Fire Temperature RGB

Colour beam	Channel	Range			Gamma
Red	IR3.9	273 0.15	350 3.2	K W m^{-2} sr^{-1} μm^{-1}	1.0 1.5
Green	NIR2.3	0 0	50 * * 12	% W m^{-2} sr^{-1} μm^{-1}	1.0 1.0
Blue	NIR1.6	0 0	50 * * 39	% W m^{-2} sr^{-1} μm^{-1}	1.0 1.0

* *The NIR2.2 and NIR1.6 calibrated values should NOT be normalized with sun zenith angle, i.e., no sun zenith angle correction be applied for these channels. In case of fires, thermal radiation is measured in NIR2.2 and NIR1.6 channels.

Day Fire RGB

Alternative names: Day Land Cloud Fire RGB, Natural Fire Colour RGB



SNPP VIIRS Day Fire RGB, 8 November 2019, 03:29 UTC, wildland fires in southeast Australia

There are two variants of this RGB, using IR3.8 or IR3.8refl in the red colour beam.

Variant 1

Main application:
- Detection of fires with smoke
- Information on vegetation, burnt area

Remarks:
- No fire intensity information
- Sensitive for moderately hot or sub-pixel fires
- Very warm surfaces may mask the fire signals (too strong red signal)

Day Fire RGB

Colour beam	Channel	Range			Gamma
Red	NIR3.8	273	333*	K	0.4
Green	NIR0.86	0	100	%	1
Blue	VIS0.64	0	100	%	1

* as with the Fire Temperature RGB, it is recommended that users in warmer, arid areas adjust the high end of the range to 343 K to avoid saturation in non-fire conditions.

Variant 2

Main application:
- Detection of fires with smoke
- Information on vegetation, burnt area

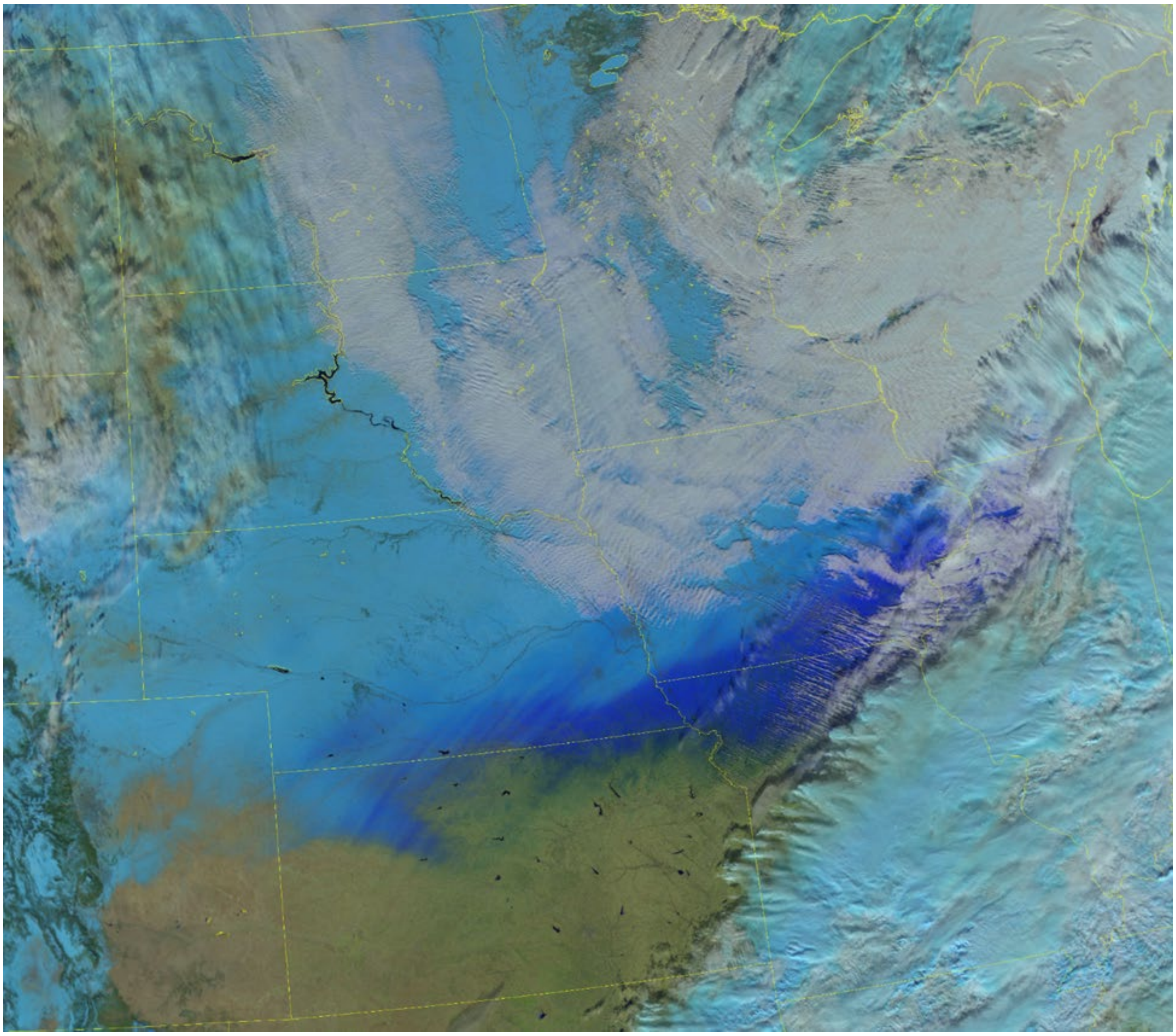
Remarks:
- No fire intensity information
- Sensitive for moderately hot or sub-pixel fires
- The background is less red for hot land surfaces than in Variant 1
- Variant 2 increases the colour contrast between actively burning and burnt or smouldering pixels over Variant 1. However, Variant 1 may be preferred for fire perimeter mapping (Seaman et al., 2023)

Day Fire RGB

Colour beam	Channel	Range			Gamma
Red	NIR3.8refl	0	50	%	1
Green	NIR0.8	0	100	%	2
Blue	VIS0.6	0	100	%	2

Day Snowmelt RGB

Alternative name: Snowmelt RGB



Main applications:
- Characterization of surface snow and ice properties
- Detection of freezing rain, sleet/mixed precipitation/ice pellet accumulations

Remarks:
- Dry snow and small-grain snow will appear light blue to cyan. Wet snow and large-grain snow will appear medium to dark blue. Freezing rain/sleet/ice pellet accumulations will appear dark blue. Otherwise, imagery has a similar appearance and interpretation as the Day Land Cloud RGB
- May be used to detect fresh snow on top of old snow and rain on snow
- As snow ages, melts, or partially melts and re-freezes, it will darken, indicating it is less likely to be lofted by high winds. As a result, it is used in conjunction with the Blowing Snow RGB to monitor for ground blizzards.

- It may be possible to detect conditions where avalanches are likely to form, however, this is often complicated by terrain- and forest-induced shadows and cloud cover, which complicate scene interpretation
- Utilizes NIR1.24, which is not currently available on any geostationary satellite

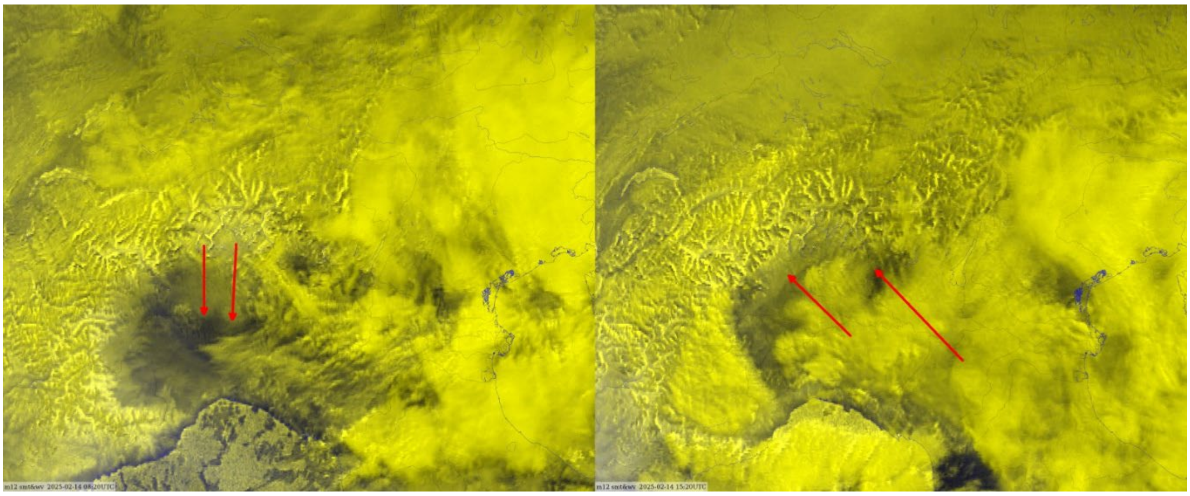
VIIRS Day Snowmelt RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.61	0	125	%	2
Green	NIR1.24	0	125	%	2
Blue	VIS0.64	0	125	%	2

RGBS UNDER DEVELOPMENT

RGBs under development: require further work and discussion, or validation. They are not recommended yet for routine production, display and training.

Day Moisture RGB



Remarks:

- New RGB with MTG FCI; from the GEO imagers, only FCI measures in the NIR0.9 band.
- It will use the combination of VIS0.8 and the new NIR0.9 channels, using a channel ratio NIR0.9/VIS0.8 (rather than channel difference)
- The term "Water Vapour Transmittance" or "Total Moisture Imagery" has been suggested for visualisations of this ratio

NIR0.91 provides information on the vertically integrated water vapour content in the atmosphere (which is usually governed by the lower levels).

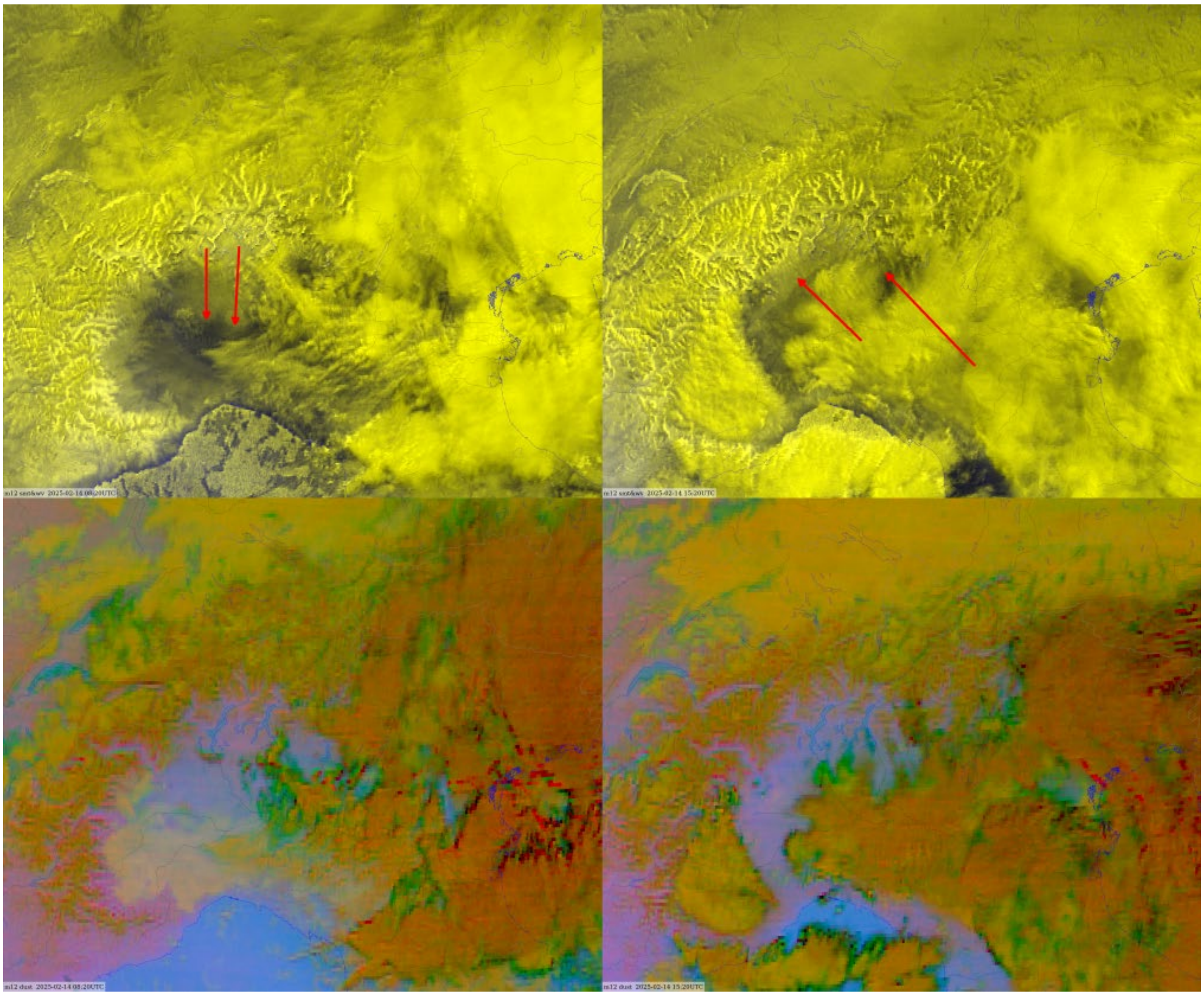
- Ref. Gao & Kaufman (2003): Water vapor retrievals using Moderate Resolution Imaging Spectroradiometer (MODIS) near-infrared channels, J. Geophys. Res., 108(D13), 4389, doi:10.1029/2002JD003023,2003

Several variants of RGBs with NIR0.9/VIS0.8 ratio are under development, testing. Neither the recipe nor the name is fixed.

On the figure below:

- To the left: foehn phase early morning – bright area spreading south
- To the right: easterly backflow (bora type) across Po Valley in the afternoon – black area under clouds pushing west
- qTPW: best watched in (fast) animation -> quasi cloud masking through human eye

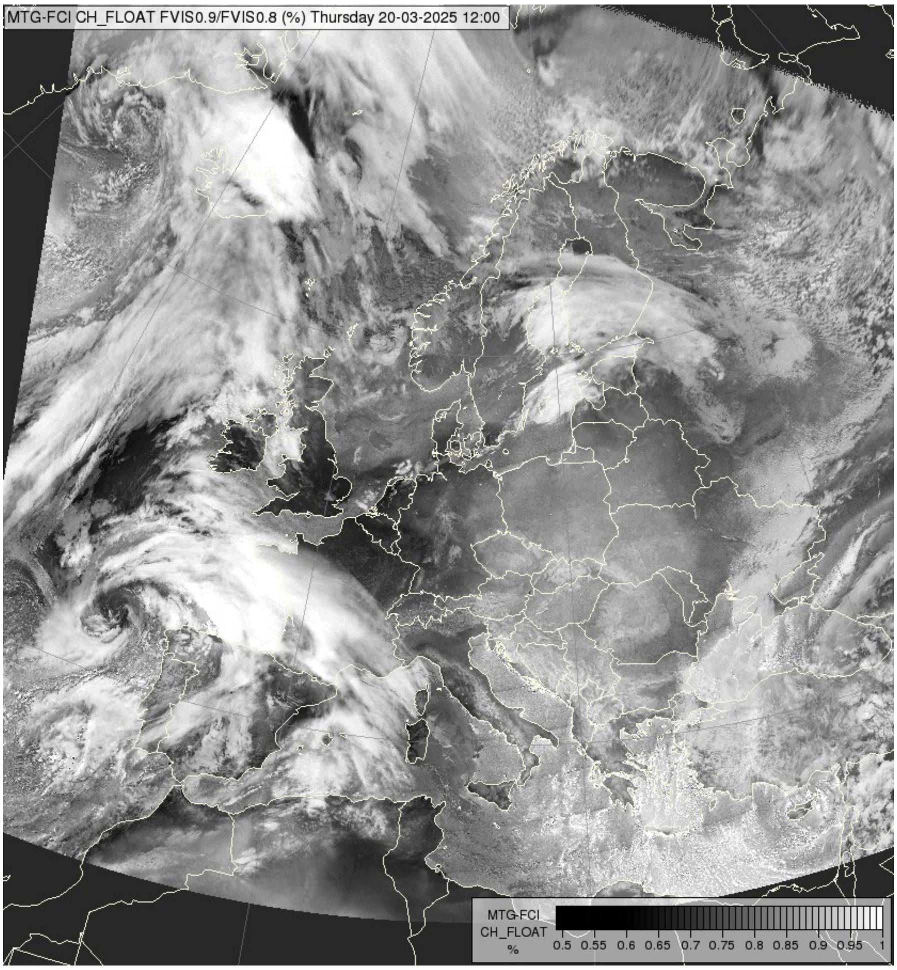
Comparison with Dust RGB - no clear sign of moisture fields:



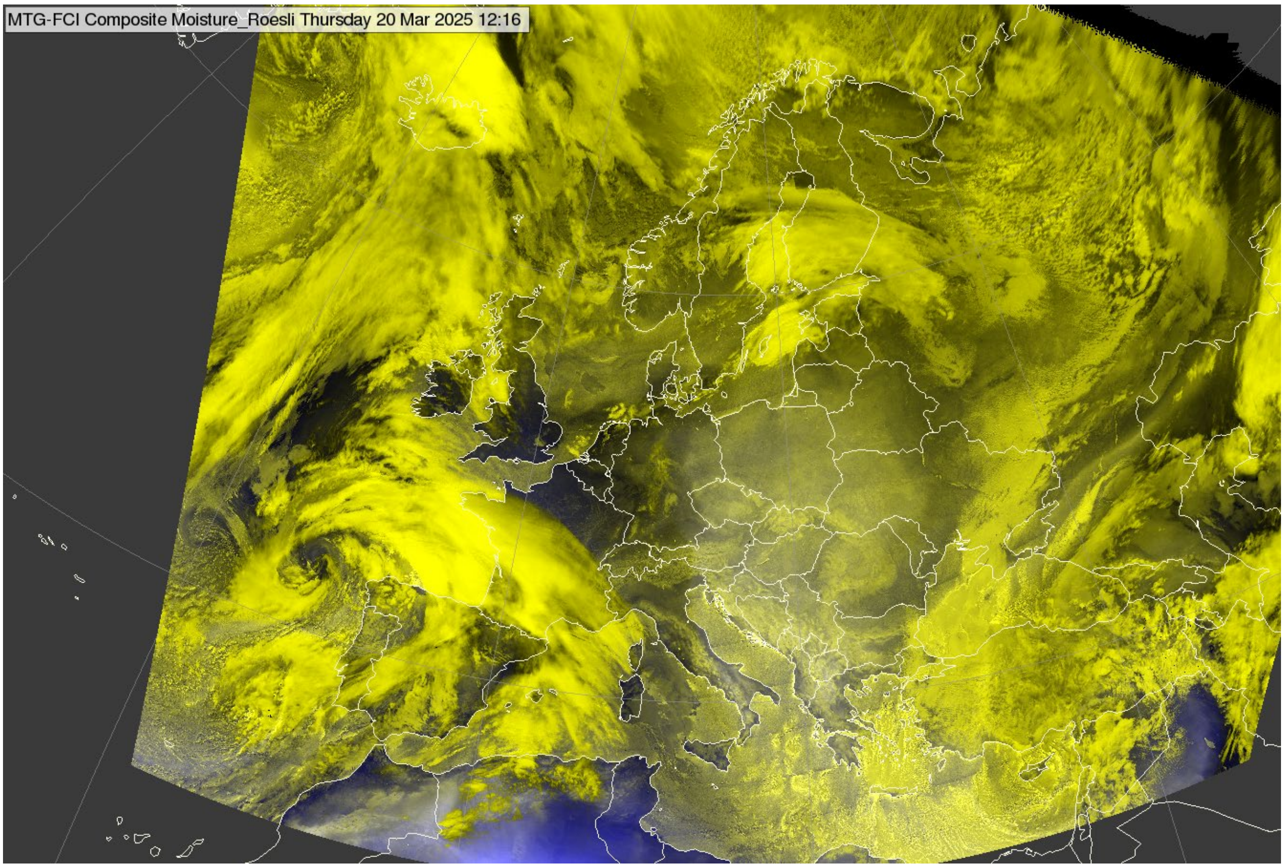
Preliminary FCI Moisture RGB (source: HP. Roesli)

Color beam	Channel (difference)	Range			Gamma
Red	NIR0.9/VIS0.8	0.6	1		0.8
Green	NIR0.9/VIS0.8	0.6	1		0.8
Blue	WV7.3	230	270	K	0.3

No solar zenith angle correction needed neither for “smt” nor for the upper RGB.



FCI smt (VIS0.9/VIS0.8) image 20 March 2025 12:00 UTC



FCI Moisture RGB, 20 March 2025, 12:00 UTC

Meteo-France tested 3 differents algorithms of moisture: 2 RGB and the ESSL visualization which is the ratio with a color palette:

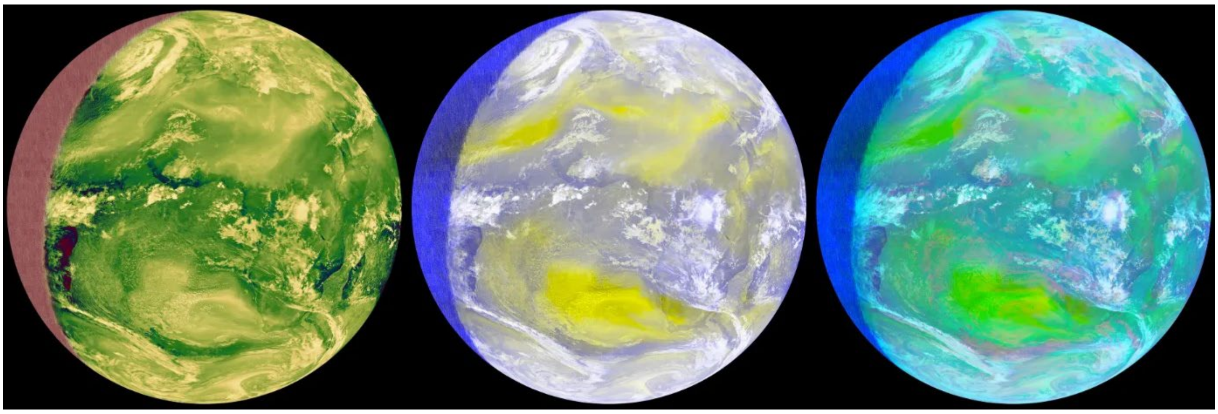


Image on the left:
NIR0.9/VIS0.8 with a color palette:

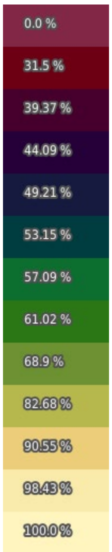


Image on the right

Color beam	Channel (difference)	Range			Gamma
Red	VIS0.6	0	100	%	1.4
Green	NIR0.9/VIS0.8	0	100	%	1
Blue	WV7.3	270	230	K	3.3

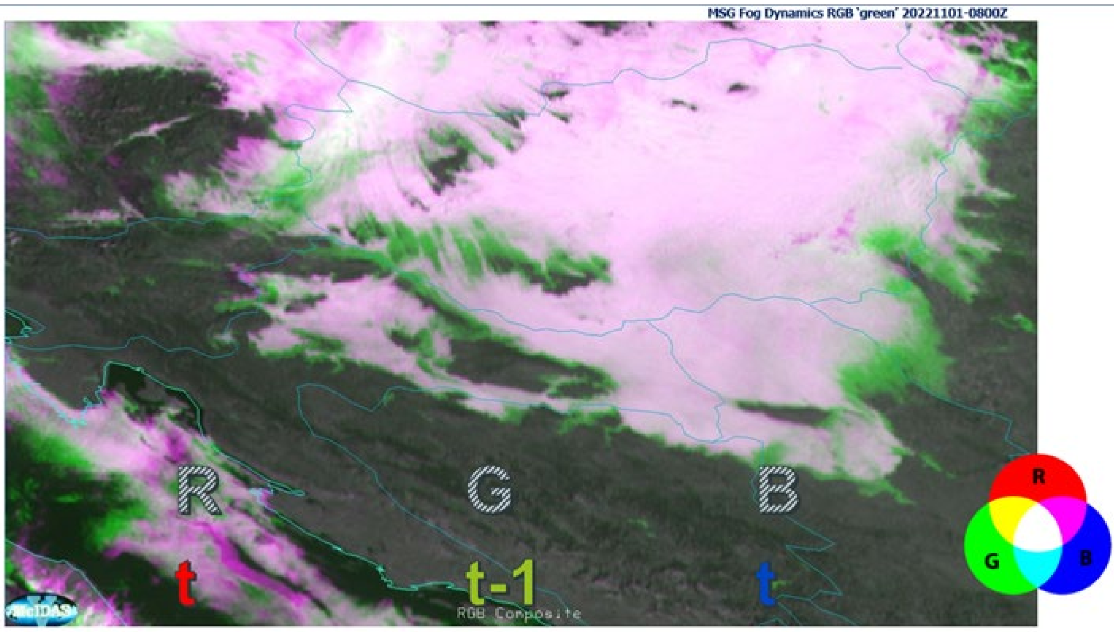
Image in the middle: HP Roesli version but not the same gamma for Blue and not sure of the range used by HP.

Color beam	Channel (difference)	Range			Gamma
Red	VIS0.9/VIS0.8	0	100	%	1
Green	NIR0.9/VIS0.8	0	100	%	1
Blue	WV7.3	270	230	K	3.3

Next steps / recommendations:

- Validate if the RGB gives additional value to a simple SMT imagery
- See if worth including additional channel (as R and G are the same contribution)
- Add cloud information (likely if some of the solar channels are included, such as NIR1.38, NIR0.91, VIS0.8 as it looks too yellow

24h Fog Dynamics RGB



Utilising high-resolution channels (eg FCI VIS0.6) and different time steps to capture changes in cloud dissipation and formation, changes in thickness.

Fog Dynamics RGB

Colour beam	Channel	Range			Gamma
Red	Hig-res VIS/NIR daytime BTD10.5-3.8 night (t)	0 -2	80 9	% K	1
Green	High-res VIS/NIR daytime BTD10.5-3.8 night (t-10 min)	0 -2	80 9	% K	1
Blue	High-res VIS/NIR daytime BTD10.5-3.8 night (t)	0 -2	80 9	% K	1

Notes:

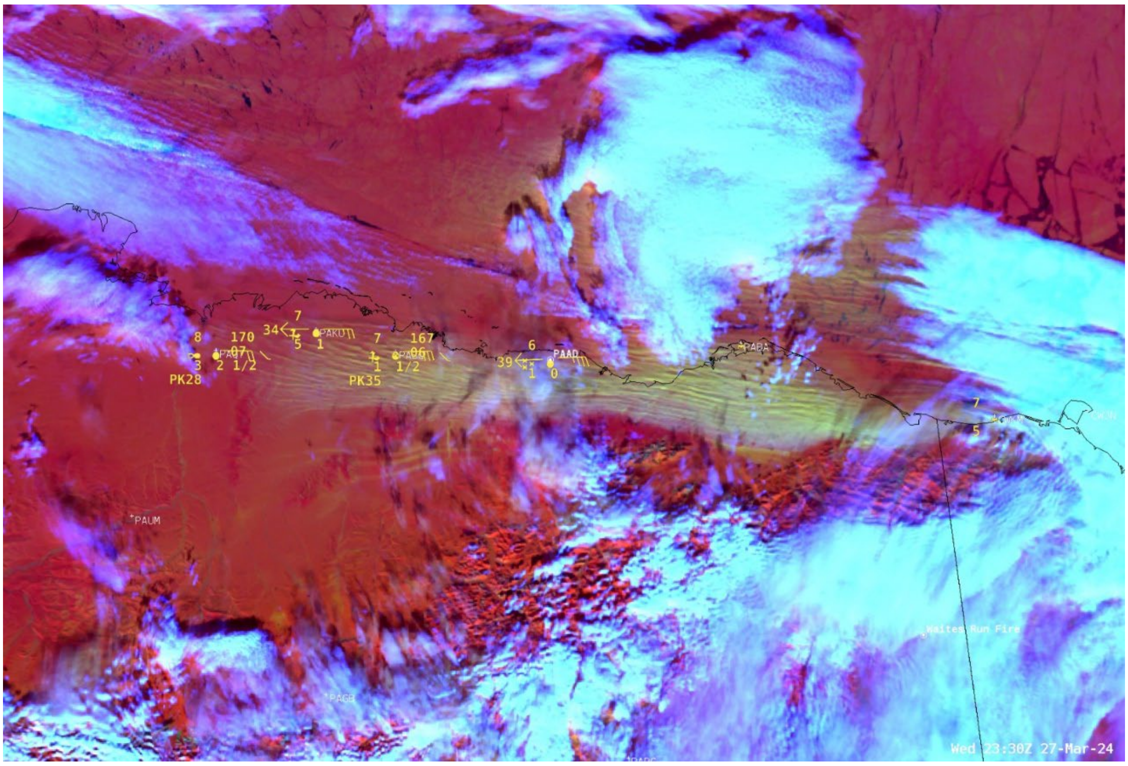
- Day/Night Fog Dynamics RGB
- Easy (almost binary) interpretation of fog formation/dissipationTime ‘t-10 min can also be’1h, 30min, 15min, X min time step? Dependent on fog dynamics?
- Convection and other purposes?
- df/dt (change) or df2/dt2 (speed of change) – Fog Speed RGB?
- Harder to implement visualisation (as uses the previous time step)

Next steps / recommendations:

- More validation required (feedback from the operational environment)
- See if Sun angle correction needed (as close to terminator image becomes biased as the two time steps have different Sun angles)

Day Blowing Snow RGB

Alternative name: Blowing Snow RGB



Main applications:

- detection of blowing snow, picked up from the ground and transported by high winds

Remarks:

- A variation of the Day Snow-Fog RGB tuned to highlight lofted snow, also known as ground blizzards
- Snow on the ground is identified in red colours. Lofted snow is identified as pink, orange or yellow, depending on plume thickness, solar illumination, and viewing geometry. Snow-free land is green.
- At night, only clouds are visible (in blue)
- Sensitive to cloud phase and particle size, however Day Cloud Phase RGB is preferred for this application
- Validated only in northern United States/southern Canada (ABI) and Alaska (VIIRS)

ABI Blowing Snow RGB

Colour beam	Channel (difference)	Range			Gamma
Red	VIS0.64	0	50	%	0.7
Green	NIR1.6	2	20	%	1.0
Blue	IR3.9 – IR10.3	0	30	K	0.7

VIIRS Blowing Snow RGB

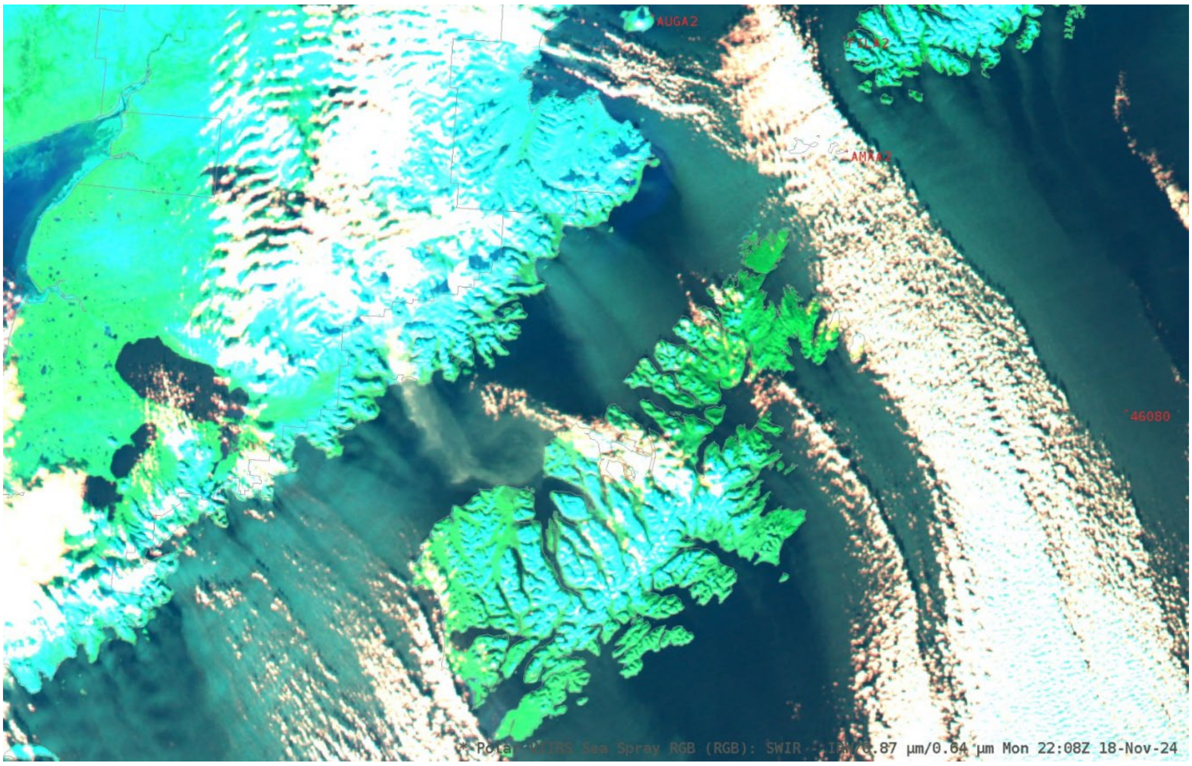
Colour beam	Channel (difference)	Range			Gamma
Red	VIS0.64	10	110	%	1.0
Green	NIR1.6	5	40	%	1.0
Blue	IR3.9 – IR10.3	0	15	K	1.0

Next steps / recommendations:

- More validation required across different sensors and different regions of the globe where blowing snow is a problem (feedback from the operational environment)

Day Sea Spray RGB

Alternative name: Sea Spray RGB



Main applications:

- detection of sea spray and other aerosols over water

Remarks:

- Sea spray can be an indicator of freezing spray, which is a significant maritime hazard in the winter
- Tuned for high latitudes, where freezing spray is expected to occur
- Sea spray is identified as light blue, cyan or gray above a background water surface (dark blue).
- Other aerosols may appear light blue, pink, gray or white, depending on optical thickness, solar illumination and viewing geometry
- Land appears light green (snow-free), light cyan (snow-covered) or white, depending on temperature. Clouds appear white.
- Sun glint is an issue
- Validated only in Gulf of Alaska (ABI and VIIRS)

ABI Sea Spray RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR3.9 - IR10.4	0	5	K	1.0
Green	NIR0.86	1	9	%	5/3
Blue	VIS0.64	2	12	%	5/3

VIIRS Sea Spray RGB

Colour beam	Channel (difference)	Range			Gamma
Red	IR3.7 - IR11.4	0	10	K	1.0
Green	NIR0.86	1	20	%	5/3
Blue	VIS0.64	2	25	%	5/3

Next steps / recommendations:

- More validation required across different sensors and different regions of the globe where sea spray is a problem (feedback from the operational environment)

OTHER VISUALISATIONS

In this section, common visualisations of satellite data that are not RGBs are listed, such as GeoColour or the Sandwich product.

GeoColour Visualisation

FCI GeoColor visualization (night part), 16 March 2025, 20:30 UTC

Main application (nighttime):

- Useful and compelling for use with general public, media

Remarks:

- GeoColour is not an RGB, but a visualisation of high interest in particular for general public and media. The GeoColour visualisation is not meant to be used in operational forecasting.
- Applicable for GOES ABI, Himawari AHI and new with MTG FCI,
- Note that this is a highly processed visualisation with many corrections applied, therefore caution is required with the interpretation
- For forecasters, at nighttime, the Night Microphysics RGB is the standard RGB for detecting fog and low clouds. Land features are sometimes visualized as low-level clouds which can be misleading.

For the nighttime part, IR channel (IR10.5) and the brightness temperature difference (IR10.5 - IR3.8) are used to identify high/mid-level and fog/low-level clouds, shown in whitish and bluish colours, respectively. They are partially transparent and used with a nighttime VIIRS-derived city lights layer.

For the daytime part, the True Colour RGB is used.

Next steps / recommendations:

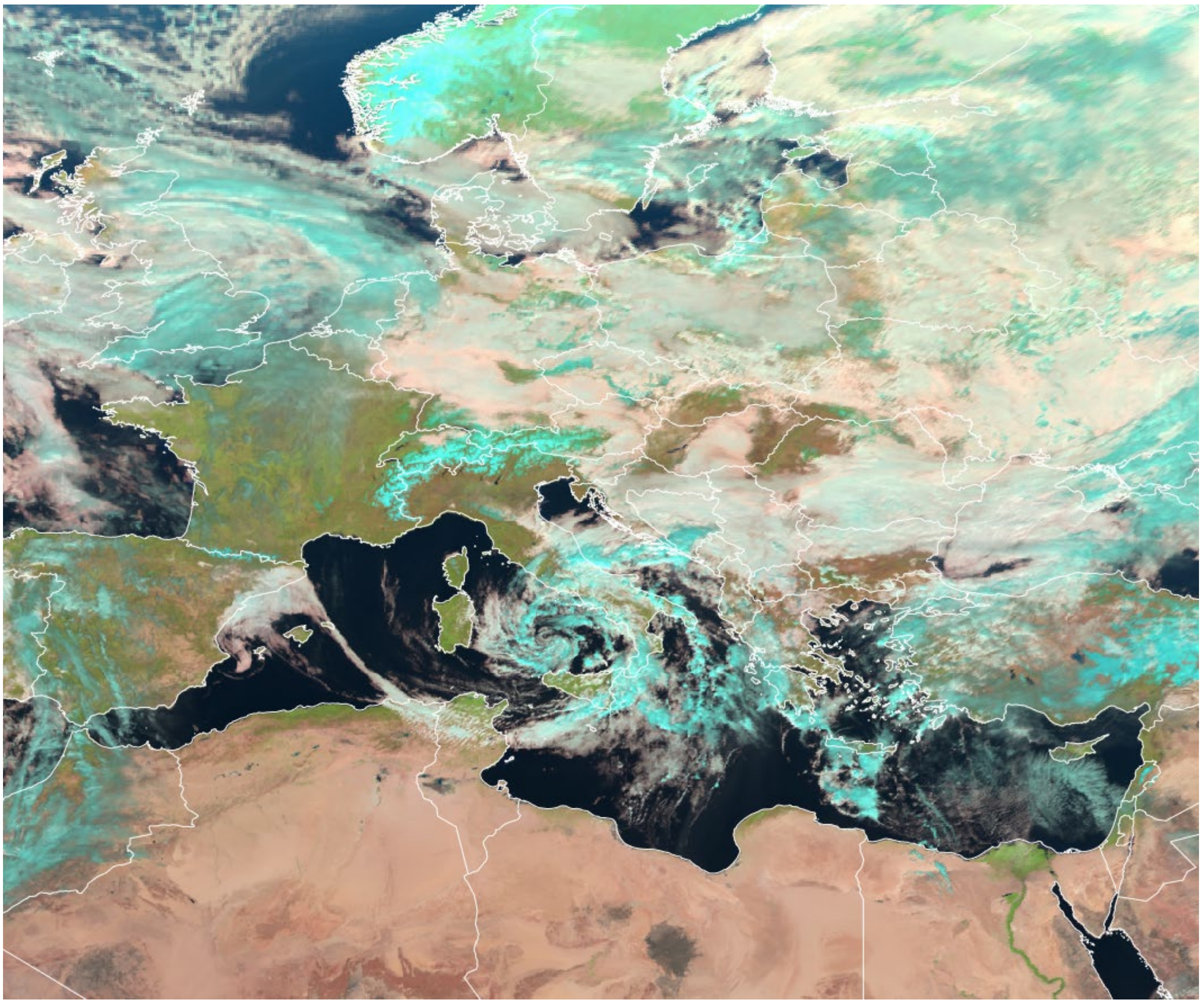
- More validation required (feedback from the operational enviornment)
- Consider using Night Microphysics RGB at night for operational forecasting purposes, or no layer (for media purposes)
- Consider including city lights as a separate layer
- See if Sun angle correction needed (as close to terminator image becomes biased as the two time steps have different Sun angles)
- GeoColour could also include lightning, fires, dust layers

LEGACY RGB COMPOSITES

This section lists RGBs that are still used operationally but have been superseded by updated or better RGBs, and that are therefore not recommended for further development.

Day Land Cloud RGB (Legacy)

Alternative name: Natural Colour RGB



SEVIRI Day Land Cloud RGB, 15 February 2025, 11:30 UTC

Main applications (daytime):

- vegetation
- cloud phase
- flood detection

SEVIRI Day Land Cloud RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.6	0	100	%	1
Green	VIS0.8	0	100	%	1
Blue	VIS0.6	0	100	%	1

MTG FCI Day Land Cloud RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.6	0	100	%	1
Green	VIS0.8	0	100	%	1
Blue	VIS0.6	0	100	%	1

GOES ABI Day Land Cloud RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.6	0	97.5	%	1.0
Green	NIR0.86	0	108.6	%	1.0
Blue	VIS0.64	0	100	%	1.0

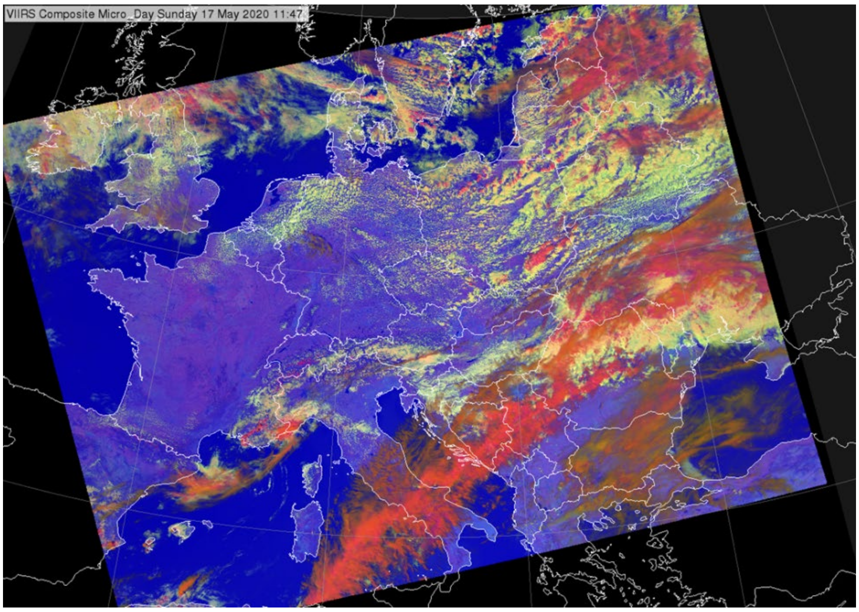
Himawari AHI Day Land Cloud RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.6	0	99	%	1
Green	NIR0.86	0	102	%	0.95
Blue	VIS0.64	0	100	%	1

FY-4 AGRI Day Land Cloud RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.61	0	100	%	1
Green	VIS0.825	0	100	%	1
Blue	VIS0.65	0	100	%	1

Day Microphysics RGB (with NIR1.6) (Legacy)



VIIRS Day Microphysics RGB (with NIR1.6) Main applications (daytime):

- General cloud analyses
- Monitoring convection
- Monitoring cloud top phase

Remarks:

- Calculation of IR3.8refl is not needed.
- It works quite well. Similar behaviour as Day Microphysics RGB, however with no inherited particle size information that is available with Day Microphysics RGB.
- Differentiation of small/large cloud top droplets is less good (for water clouds). It is sensitive to cloud top particle size in somewhat narrower range than the Day Microphysics RGB is.
- This recipe is used for Metop AVHRR and VIIRS images, and with SEVIRI.
- AVHRR on some satellites swaps between 1.6 during the day and 3.7 during the night, so the normal day microphysics cannot be generated during the day.

Colour beam	Channel	Range			Gamma
Red	VIS0.8	0	100	%	1
Green	NIR1.6	0	70	%	1
Blue	IR10.5	203	323	K	1

Day HRV Cloud RGB (Legacy)

Main application
- Monitoring convectionMonitoring clouds with high resolution imagery

Remarks:
- Very old RGB type
- Nice images, easy to understand
- This RGB reflects cloud optical thickness and cloud top temperature information.
- In case of SEVIRI, HRV was the only high-resolution channel. That is why it is used in two colour beams. With FCI all shortwave channels have higher resolution than the IR channels. No need any more to visualise the same shortwave channel in two colour beams.

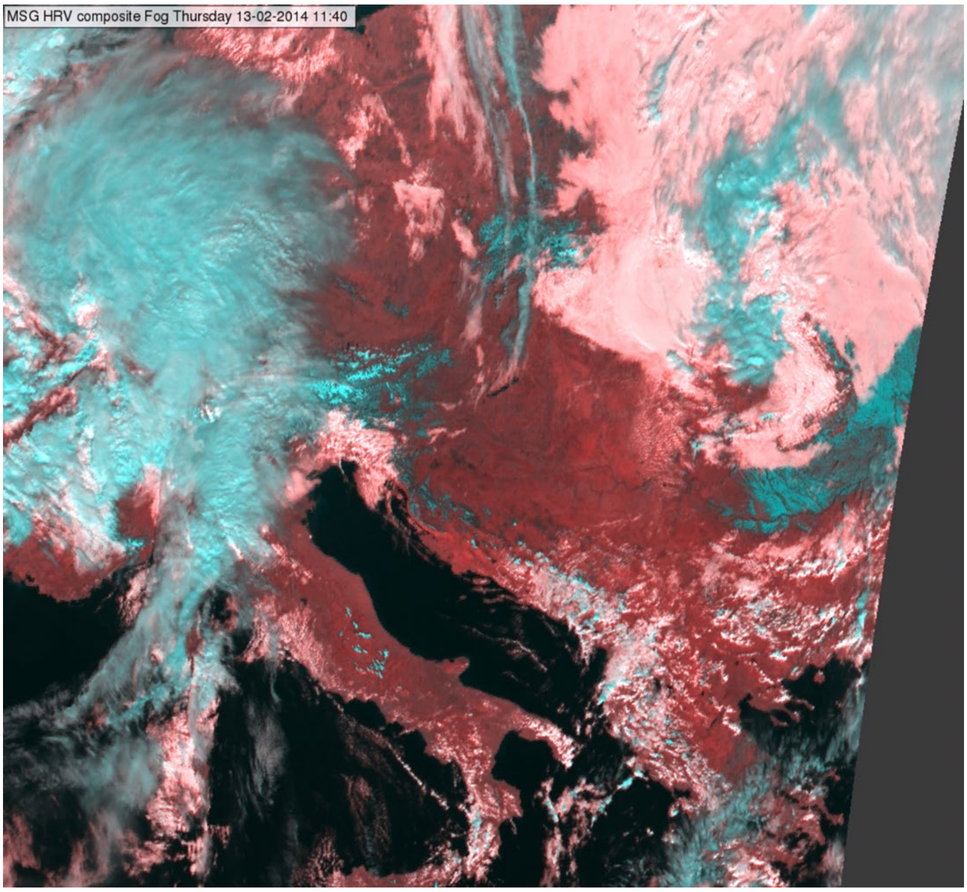
SEVIRI HRV Clouds RGB

Colour beam	Channel	Range			Gamma
Red	HRV	0	100	%	1
Green	HRV	0	100	%	1
Blue	IR10.8	323	203	K	1

Day Cloud Convection RGB

Colour beam	Channel	Range			Gamma
Red	VIS0.64	0	100	%	1.7
Green	VIS0.64	0	100	%	1.7
Blue	IR10.3	323	203	K	1.0

HRV Fog RGB (Legacy)



SEVIRI HRV Fog RGB, 13 February 2014, 11:40 UTC

Main application
- Monitoring snow covered land and fog/low cloud in high resolution
- Monitoring clouds in general (includes a cloud phase and thickness information)

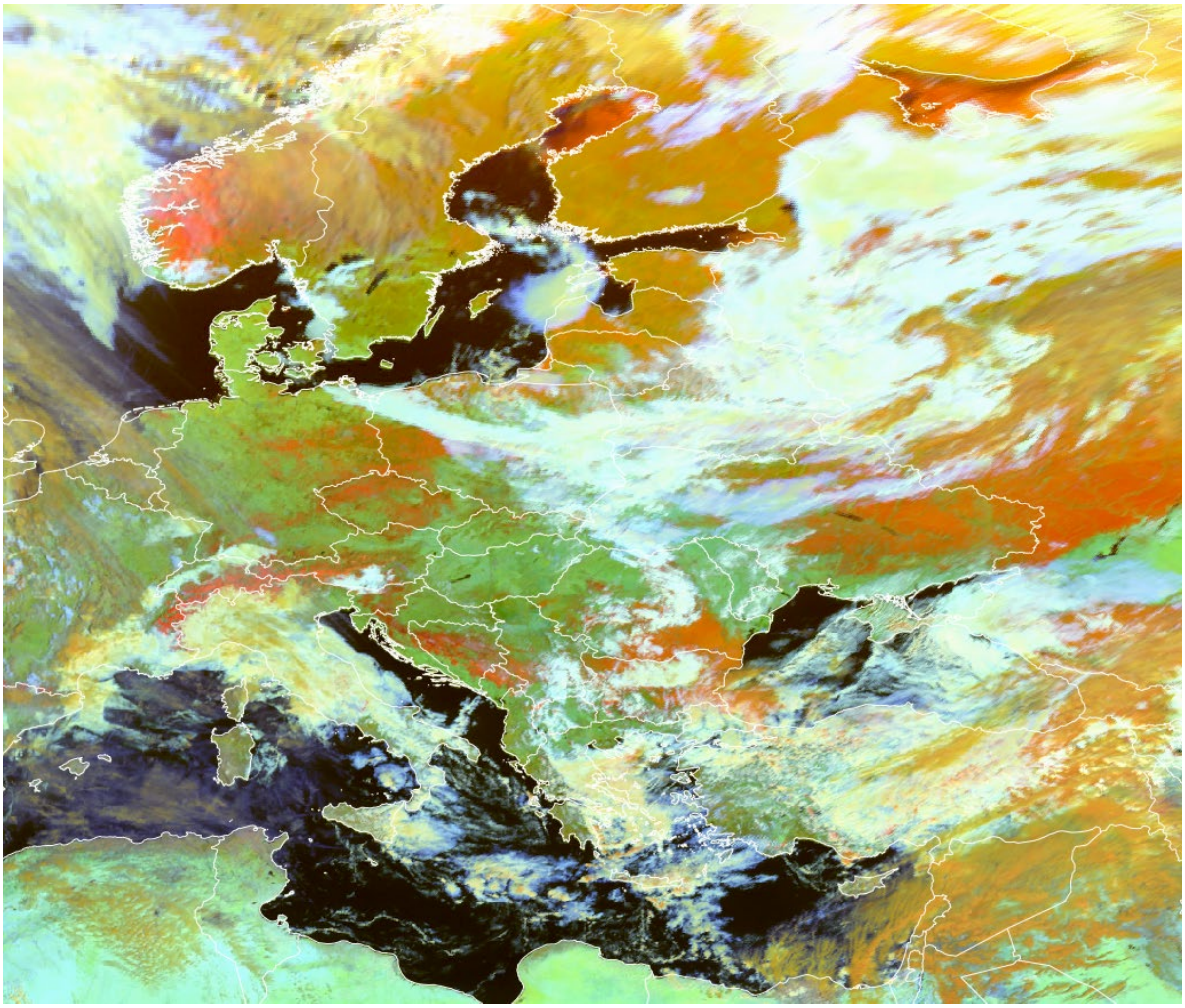
Remarks:
- Nice images, easy to understand
- In case of SEVIRI, HRV was the only high-resolution channel, which is why it is used in two colour beams. With FCI, all shortwave channels have higher resolution than the IR channels. No need any more to visualise the same shortwave channel in two colour beams.
- Superseded by Cloud Phase RGB and Cloud Type RGBs for detecting snow and low clouds

HRV Fog RGB

Colour beam	Channel	Range			Gamma
Red	NIR1.6	0	70	%	1
Green	HRV	0	100	%	1
Blue	HRV	0	100	%	1

Day Snow-Fog RGB (Legacy)

Alternative name: Snow RGB



SEVIRI Snow RGB, 19 February 2025, 10:30 UTC

Main applications:

- snow detection
- Good colour contrast between fog/stratus and snow

Remarks:

- For snow detection, Cloud Phase RGB or Cloud Type RGB are also excellent. Also using two microphysical NIR channels where snow reflection is low.
- Additionally, for clouds, the performance of Cloud Phase RGB is better than for Snow RGB.

MSG SEVIRI Day Snow-Fog RGB

Colour beam	Channel	Range			Gamma
Red	VIS0.8	0	100	%	1.7
Green	NIR1.6	0	70	%	1.7
Blue	IR3.9refl	0	30	%	1.7

GOES ABI Day Snow-Fog RGB

Colour beam	Channel (difference)	Range			Gamma
Red	NIR0.86	0	100	%	1.7
Green	NIR1.6	0	70	%	1.7
Blue	IR3.9 – IR10.3	0	30	K	1.7

Himawari AHI Day Snow-Fog RGB

Colour beam	Channel (difference)	Range			Gamma
Red	NIR0.86	0	102	%	1.6
Green	NIR1.6	0	68	%	1.7
Blue	IR3.9 refl	2	45	%	1.95

FY4 AGRI Snow RGB

Colour beam	Channel (difference)	Range			Gamma
Red	NIR0.825	0	100	%	1.7
Green	NIR1.6	0	60	%	1.7
Blue	IR3.75 refl	0	30	%	1.7

RGB NAMING CONVENTIONS [2]

Full name	Alternative / short name	Spanish translation	German translation
Basic RGBs			
24-hour Microphysics Cloud RGB	24-h Microphysics RGB	RGB Microfisica de nubes 24h	24h Mikrophysik RGB
Night Microphysics RGB	Fog RGB, Night Fog RGB, Fog/Low Cloud RGB, Fog/Low Stratus RGB	RGB Niebla	Nacht Mikrophysik-RGB
Day Cloud Phase RGB	Cloud Phase RGB	RGB Fase de nube	Wolkenphase RGB
Day Cloud Type RGB	Cloud Type RGB	RGB Tipo de nube	Wolkentyp-RGB
Special Applications RGBs			
24-hour Microphysics Dust RGB	Dust RGB	RGB Polvo	Staub RGB
24-hour Microphysics CVD Dust RGB	CVD Dust RGB		
24-hour Microphysics Ash RGB	Ash RGB	RGB Cenizas volcanicas	Asche RGB
Day Severe Storms RGB	Convection RGB, Severe Storms RGB, Severe Convection RGB	RGB Tormentas Severas	Schwere Konvektion-RGB
Volcanic Emissions (SO2 and Ash) RGB	SO2 and Ash RGB		
Day Fire RGB	Day Land Cloud Fire RGB, Natural Fire Colour RGB		
Day Snowmelt RGB	Snowmelt RGB		
Legacy RGBs			
Day Land Cloud RGB	Natural Colour RGB		
Day Snow-Fog RGB	Snow RGB		

[2] No proposed translation into French, Arabic since Meteo-France and CoE Oman recommend using the English terminology